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REVIEW

Application of artificial intelligence for asset integrity management of offshore oil and gas pipelines

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Abstract

The assets and infrastructure of offshore oil and gas platforms, such as pipelines, operate in harsh environments and often require frequent maintenance for sustained operation. The conventional asset management protocols, such as pigging, aerial and surface inspection, and ultrasonic testing, are resource-intensive and often reactive, leading to high operational costs and unplanned downtime. In this context, the capabilities of artificial intelligence (AI) can be effectively leveraged to manage these assets efficiently and economically. This paper investigates the recent advances in AI-driven asset management to enhance the reliability, safety, and cost-effectiveness of offshore oil and gas pipeline asset management. The study begins with mapping recent research on AI applications in oil and gas pipeline management using citational network analysis. Eight key thematic clusters were identified, each highlighting specific research areas and future directions. Based on these insights, a structured AI-based asset integrity management system is proposed for oil and gas infrastructure, including pipelines, accounting for practical implementation considerations such as system integration, data reliability, and regional constraints. The system is structured into three phases: operational asset integrity management, asset life extension, and AI-driven applications. This approach integrates both qualitative and quantitative methods, utilizing real-time data analytics, machine learning, and predictive modeling. The proposed protocols are designed to be a practical and affordable roadmap for achieving continuous, reliable and safe integrity management supported by real-time analytical and economic justification. While the framework offers broader applicability, regional environmental factors and data variability may require adjustments to suit specific contexts.

Keywords Asset integrity · Artificial intelligence · Oil and gas · Offshore assets · Fabric maintenance · Autonomous integrity management

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1 Introduction

Safe operations of oil and gas assets are crucial for the profitability of oil-producing companies. Historically, catastrophic disasters such as the Piper Alpha rig incident in 1988 and the oil spill at the Deepwater Horizon site in 2010 underscore the detrimental effects of inadequate asset management (Mcclean Millar 2015). These events highlight the pressing need for a comprehensive management system that ensures process safety and effectively mitigates risks. Operational integrity is crucial in managing risks to As Low As Reasonably Practicable (ALARP) while prioritizing the safety of workers and equipment, necessitating the establishment of an effective asset integrity management program.

Adoption of advanced technologies is not just beneficial but essential. Advanced tools enhance decision-making,

streamline operations, and optimize overall performance. Asset integrity management (AIM) is set to be revolutionized by automation and improved asset maintenance strategies, enabling businesses to achieve their desired outcomes more efficiently.

AIM involves the adoption of a systematic approach to ensure that equipment functions safely throughout its entire lifespan, a concern that becomes increasingly urgent as equipment ages (Chandima Ratnayake and Markeset 2012). The British Standards Institution (BSI), in its publication "Strategic Asset Management PAS 55:2008", recommends a holistic approach to strategic asset management, which is further elaborated in ISO 55000 (British Standards Institution 2014). According to Adegbite et al. (2023) the emergence of Artificial Intelligence (AI) and Machine Learning has brought about significant changes in maintenance strategies of critical infrastructures through the use of predictive maintenance and decision making in the oil and gas industry (Edo et al. 2024). Besides enhancing reliability in the systems, these technologies also help to eliminate several risks that are often associated with operations. The Oil and gas industry plays a major role in supporting the economy and development of many countries around the world, so its sustainability and efficiency are critical (Usiagu et al. 2024).

For operating companies having operating fields located offshore, maintaining the asset integrity of offshore installations is a labor-intensive process. Traditionally, this is done by frequent site visits by boat or helicopter under challenging climatic conditions. Integrity engineers manually review collected data and assign maintenance tasks. Preventive maintenance is carried out at fixed time intervals, and in some instances, this may be performed unnecessarily on equipment that is still functional and contributes to the costs of maintenance and time wastage (Usiagu 2024). Transitioning from conventional to automated practices based on actual condition and risk demands the integration of emerging technologies, particularly those that substitute manual inspection and monitoring methods with automated systems. The implementation of advanced condition-monitoring sensors allows for data to be remotely relayed to a central integrity management system, where it can be analyzed, and decisions can be made with minimal human intervention. Although the initial capital investment for these technologies is high, they offer long-term benefits by reducing logistical costs and minimizing the risk associated with offshore travel. An example is AeroX (Trujillo et al. 2019), which is an airborne robot manipulator capable of conducting physical contact tests.

Corrosion in pressure equipment such as piping and pressure vessels is one of the major challenges in managing the integrity of offshore oil and gas assets. Internal and external corrosion reduces the wall thickness of these components, compromising their ability to withstand operating pressures. If not addressed in time, this can lead to leaks, ruptures, or even explosions, thereby posing serious risks to personnel, infrastructure, and the environment. These risks are even more critical for ageing oil and gas platforms that continue to operate beyond their original design life.

Modern technologies now enable faster and more accurate data analysis, facilitating pattern recognition and model-based decision-making to trigger timely responses for managing pipeline infrastructure. This automation minimizes human intervention, simplifies complex tasks, and saves both time and effort. Furthermore, the availability of cost-effective cloud computing solutions provides an efficient means for storing and transmitting inspection data, ensuring real-time access from any location. By enabling automated, condition-based maintenance, these technologies reduce the need for manual inspections and improve asset management through faster, more standardized maintenance planning. In view of the above, a study has been conducted with the following dual objectives:

1. **To comprehend the current state of the field**: This study uses Citation Network Analysis (CNA) to examine over 177 publications on AI in the management of oil and gas pipeline infrastructures. Modularity-based clustering revealed eight thematic research clusters, with each publication averaging four citation links. The analysis mapped key evolutionary paths, highlighted emerging research fronts and challenges, and exposed critical gaps, particularly in technical and economic feasibility, legislative frameworks, and standardization of AI in AIM of oil and gas pipelines.
2. **To develop and substantiate a three-stage AI-based framework for AIM**: The framework will address the systematic identification and mitigation of integrity risks to a level that is As Low As Reasonably Practicable (ALARP). It will also define and manage critical factors influencing asset longevity, particularly for pipeline assets operating beyond their design life, to support informed decision-making while maintaining safety and performance. Additionally, the framework will integrate AI-based technologies for enhancing AIM through predictive maintenance, real-time monitoring, and intelligent decision support.

The contributions of this paper are as follows:

- In contrast to previous reviews in this area, the present study employs Citation Network Analysis (CNA) to systematically map the research landscape. This approach enables a clearer understanding of how research concepts have evolved over time, helps identify knowledge gaps, and highlights promising areas for future investigation.

- In contrast with the generic AIM strategies followed by the industry, the proposed AI-based AIM is specifically tailored for oil and gas platforms, working beyond their expected design life period.
- We propose a novel Multi-Modal AI approach for comprehensive management of both internal and external corrosion. The system deploys sensor technology for monitoring internal corrosion and utilizes image analysis techniques to classify external corrosion and initiate appropriate maintenance actions, enabling an integrated and intelligent integrity monitoring solution.
- By proposing the use of Explainable AI, the effect of key environmental factors such as pressure and temperature on wall thickness reduction can be identified. This empowers maintenance teams to prioritize critical parameters, optimize resource allocation, and plan interventions more effectively.

The remainder of this paper is structured as follows: Sect. 2 reviews previous studies on Asset Integrity Management (AIM) in oil and gas pipelines, using citation network analysis, and provides an overview of fundamental asset integrity practices. Section 3 describes the research methodology and introduces the proposed three-stage AIM framework, detailing the AI tools and techniques employed for condition monitoring, predictive maintenance, and asset life extension. This section also explains how AI is integrated into inspection processes and decision-making tasks. Finally, Sect. 4 concludes the study and outlines future research directions, highlighting the importance of continued exploration of AI integration in offshore integrity management.

2 Review of artificial intelligence in oil and gas pipeline assets management

2.1 Citation network analysis-bibliometric analysis

In this study, we perform a main path analysis of a citation network focusing on the application of artificial intelligence (AI) in asset integrity management within the pressure equipment infrastructure of the oil and gas sector, based on research articles retrieved from the Web of Science (WoS). As illustrated in Fig. 1, the citation network was constructed by collecting relevant citation data via WoS.

The initial step in the main path analysis involves assigning a weight to each directed link between two nodes in the citation network. To assess these link weights, three key metrics are commonly used, which are Node Pair Projection Count (NPPC), Search Path Node Pair (SPNP), and Search Path Link Count (SPLC) (Gustafsson et al. 2014). Among these, SPLC is considered the most important, as it reflects the number of search paths that pass through a specific link, indicating its significance in the overall knowledge flow (Batagelj 2003). A higher SPLC value implies that more citation paths traverse that link.

To construct the main path, the algorithm starts at a source node and follows the heaviest weighted links (based on SPLC) until a sink node is reached. Each link along the main path represents a critical knowledge transmission channel, highlighting the progression and development of scientific understanding. These main path nodes serve as key milestones within the research domain (Garfield 1964). Although some search methods prioritize direct connections

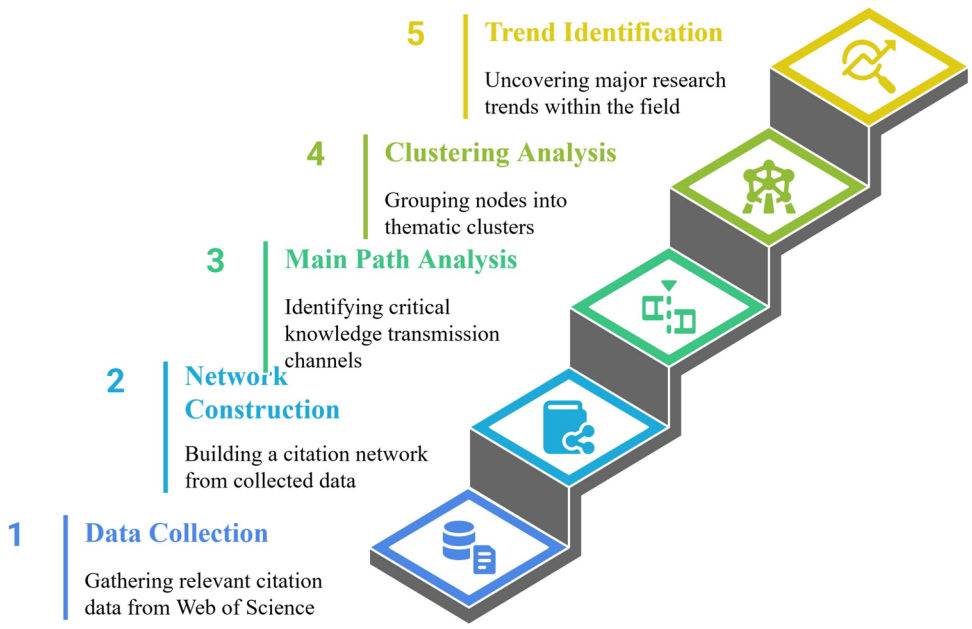


Fig. 1 Schematic representation of the main path analysis

with high traversal counts, they may not always capture the most comprehensive view of the citation flow. In this analysis, the global main path approach was used, wherein the initial node is identified, and links are successively added based on maximum traversal counts until the sink node is reached. We implemented this main path analysis using the GEPHI software (Sivaprasad et al. 2020). The citation data obtained using keywords such as "internal corrosion," "external corrosion," "AI," and "asset integrity management in oil and gas pipelines." The resulting citation network comprised 765 nodes, including cited and citing publications (Gustafsson et al. 2014). To uncover the major research trends, we conducted a modularity-based clustering analysis on the citation network (Mclean et al. 2016) and the number of nodes reduced to 177 with a minimum citation as 4, which yielded eight major clusters as detailed in Table 1, each representing a distinct thematic area within the field.

These clusters were further analyzed individually to trace their respective key evolutionary routes, each marked by milestone papers. As shown in Fig. 2, the evolution of main path analysis for each cluster highlights the steps that lead to content analysis from data extraction, network construction, and main path identification. The clusters derived from the analysis are discussed in the following sections.

Table 1 Summary of the eight identified clusters

S. No.	Cluster	Theme	Contribution to AIM
1	Pipeline corrosion	AI-driven corrosion prediction and management using sensor data and machine learning	Enables predictive corrosion maintenance and reduces unplanned failures
2	Structural integrity	Predicting degradation and structural risks in offshore pipelines using hybrid AI-based physics models	Improving structural resilience by forecasting critical degradation zones
3	Failure analysis	Analysing and forecasting failure mechanisms using AI-based pattern recognition	Supports proactive failure prevention and maintenance planning
4	Adaptive integrity assessment and management	Provide dynamic integrity strategies based on real time risk and adaptive corrosion allowance	Helps dynamic real-time asset management
5	Probabilistic and stress-based modelling	AI integrated probabilistic failure modelling for real time decision support	Enhances risk-based decision making under uncertain conditions
6	Pitting corrosion	Hybrid AI models are used for statistical and simulation based analysis of pitting corrosion	Improves the failure prediction of localised pitting
7	Dented pipelines	FEM and deep learning is used in stress and deformation assessment of dented pipelines	Helps in early detection and prioritisation of dent defects
8	Pipeline system integrity	Management of system-based integrity and reliability using Bayesian and AI Models	Enables holistic integrity and reliability across pipeline networks

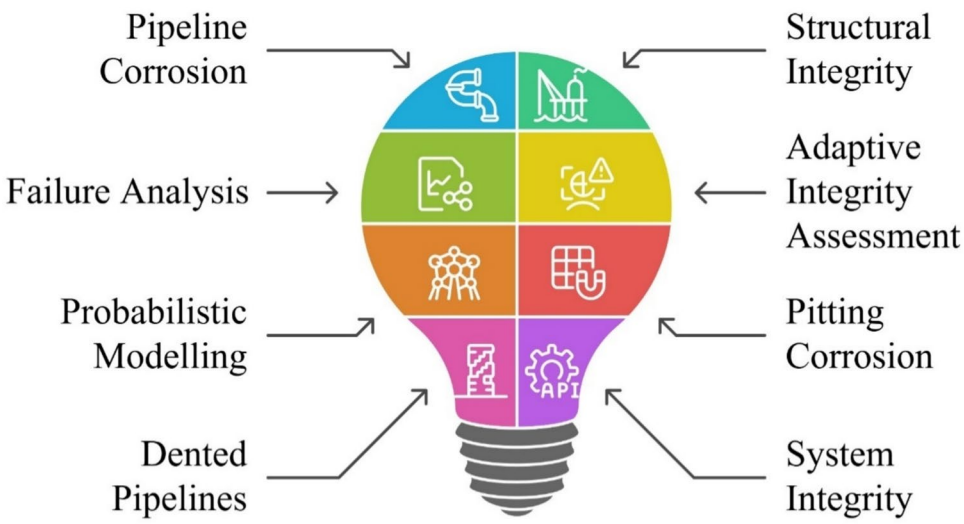


Fig. 2 Streams of AI in pipeline integrity management

2.2 Cluster 1 pipeline corrosion

AI integration transforms pipeline corrosion monitoring and mitigation. The main advantage AI offers is its capability to explore large datasets, spot the hidden patterns and generate real-time risk assessments. With machine learning models, specifically deep learning architectures, the past corrosion patterns and sensor data can be learned to provide early warnings of potential failures. Also, the fusion of AI with real-time industrial IoT (IIoT) sensors provides a proactive maintenance strategy where predictive analytics helps decision-making before reaching a critical threshold. Dynamic optimization of maintenance schedules based on operational costs and risk mitigation can also be performed by reinforcement learning. Digital twins based on AI can be used for simulating different corrosion scenarios in a virtual environment before embarking on intervention methods in the field. Figure 3 schematically shows the application of AI in pipeline Corrosion Prediction and management.

The contributions of the studies in this cluster are important in this field as they reveal new directions on predictive modelling for corrosion and the broad application of AI in pipeline maintenance. For example, deep learning has been used to make a significant advancement in predictive modelling of corrosion in buried transmission pipelines, as shown in (Akhlaghi et al. 2023).

Unlike static datasets and simple empirical formulas, deep neural networks are used in the study to analyze big corrosion data and uncover complex relationships among corrosion conditions and pitting depth development. The study showed that machine learning can accurately predict the failure of pipelines by using a predictive AI framework. AI can process huge and complicated datasets in real time.

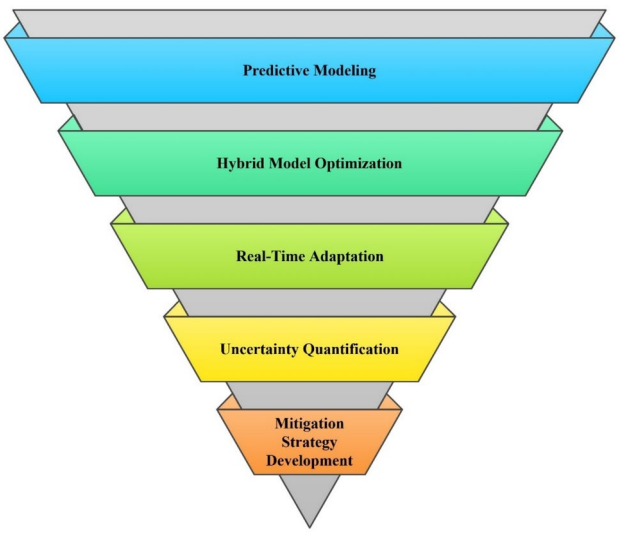


Fig. 3 AI in pipeline corrosion prediction and management

This makes it suitable for real-time monitoring and predictive maintenance compared to traditional statistical models.

Theoretical modelling of crevice and corrosion has been explored, focusing on the electrochemical processes involved in localized corrosion behaviour in (Sharland 1987). This work points out the limits of deterministic models in describing the randomness and variability of pitting initiation and growth. These findings inspired an AI-based, probabilistic approach that adapts to different pipeline conditions instead of the rigid, one-size-fits-all formula. Corrosion models could have an increased predictive power using current modern AI methods, specifically Bayesian neural networks and probabilistic deep learning, which can estimate the uncertainty in their forecasts.

An AI-driven model, utilizing predictive analysis to estimate the maximum corrosion depth in oil and gas pipelines, has also been developed for corrosion prediction (El Amine Ben Seghier et al. 2020). Using a combination of genetic algorithms and machine learning techniques, the study adopts predictive accuracy optimization via combined model parameter tuning. Consequently, this research highlights the value of the hybrid AI models expressed by the intersection of data-driven approaches with optimization methods to learn adaptively and over time. For instance, such enhanced AI models can be applied in the real-world pipeline networks, where operators will be able to dynamically adjust maintenance schedules contingent upon an always updated corrosion forecast. This results in a significant reduction in operational costs and downtime.

Another study to predict the internal corrosion in oil and gas pipelines through predictive models is described in (Papavinasam et al. 2010). Their model integrates chemical and environmental variables in predictive equations, bridging the gap between the laboratory findings and real field applications. However, the inherent deterministic nature of the model limits its applications in nonlinear corrosion problems under fluctuating conditions. This approach could be made more sophisticated with AI-driven predictive models using reinforcement learning, which are based on real-time monitored data. Such models can learn continually from the new data and thus can offer an accurate as well as an adaptable corrosion prediction.

The stochastic nature of corrosion has been presented in (Valor et al. 2013), where the corrosion progression is modelled using Markov chain. This approach efficiently captures the nonlinear process of corrosion, which has probabilistic transitions between the stages of degradation. The limitation of this approach is that, for practical applications, a significant amount of historical failure data is required. A better approach under such a situation is to integrate Markov-based probabilistic models with machine learning, especially in predicting the corrosion probabilities at various time intervals.

Despite the recent initiatives for AI-driven corrosion prediction in AIMS, there remain several important areas that need further research and development. A major limitation of current studies is the insufficient adaptability of the proposed AI models. Current machine learning and deep learning models are trained on historical corrosion data, but they do not continually learn from current sensor inputs. Future research should focus on developing self-learning AI models that can dynamically enhance their predictive capabilities by continuously learning from real-time data collected via industrial IoT (IIoT) sensors, including pipeline operating conditions such as pressure, temperature, and environmental variations. This would continuously improve the corrosion predictions and dispatch maintenance schedules.

In addition, it is also necessary to quantify the uncertainties in the AI-based corrosion predictions. Corrosion depth estimations and risk assessments are currently included in the existing AI models, but they do not provide the level of confidence in their predictions. In decision-making in high-risk industries such as oil and gas, engineers need models to inform them whether there is prediction uncertainty to make possibly more effective risk assessments and management. Deep learning and probabilistic AI approaches could be integrated into Bayesian deep learning to quantify uncertainties of corrosion forecasts. Most of the existing AI-based corrosion models predict the pipeline material degradation under standard conditions, whereas real-life pipelines are exposed to a wide range of interrelated environmental conditions. Future studies should focus on such multi-environmental AI models for pipelines which work under diverse conditions of offshore, deep sea, high altitude and desert environments. More site-specific corrosion predictions could be derived by incorporating further inputs on soil composition, humidity, salinity, and microbial activity-related environmental parameters in the AI models. Finally, hybrid models consisting of electrochemical simulation and AI-driven pattern recognition may increase the predictive accuracy over a range of environmental conditions.

At present, AI models are highly effective at forecasting the corrosion progress, but not in proposing the best intervention techniques. Hence, along with the corrosion forecasting, future research should be extended to possible AI-driven corrosion mitigation strategies as well. This can be achieved through reinforcement learning, which can suggest the most fruitful mitigation strategies, such as coating modification, cathodic protection, decaying or altering inhibitor injection rates. These models can be developed based on the past maintenance actions and their outcomes, giving recommendations for proactive corrosion control.

Several AI-based corrosion models have been validated in controlled studies or on small datasets. Their performance across an entire pipeline system, under the real operational environments, is yet to be verified. Hence, the scalability and deployability of AI-based corrosion models in large-scale pipe networks are also a future research priority. Future work should also be focused on cloud-based AI platforms where the corrosion predictions can be combined with existing asset management systems to enable easy deployment and operational scalability. Similarly, distributed AI models which can work on data from multiple pipelines at the same time are also required.

2.3 Cluster 2 structural integrity

Corrosion and instability in steel pipelines, as well as seismic-induced deterioration in steel frames, are prevalent civil and structural engineering issues. The traditional analytical and experimental approaches can give a basic understanding on the material degradation and failure mechanisms. However, they fail in fully explain the phenomena and their impact on existing and future design concepts.

The problem of corrosion in the steel pipelines, particularly those subjected to a combined axial compression and internal pressure, remains a critical problem for the oil and gas industry. Corrosion-induced wall thinning causes a reduction of axial strength, whilst it can simultaneously enhance ductility, making the relationship between them structurally complex. The former was empirically substantiated in earlier works, showing that corrosion depth inversely affects load carrying capacity and promotes local buckling failure modes under axial deformation. However, due to the lack of predictive tools in these studies, they are not useful in real-life applications where proactive maintenance is necessary (Dewanbabee and Das 2013).

A novel hybrid deep learning model termed as the Convolutional Gated Recurrent Unit (CGRU), presented in (Seghier Ben et al. 2025), aims to predict atmospheric corrosion of steel structures subjected to a maritime environment. The model learns such spatial features (salt-induced conductivity), as well as temporal trends (humidity and temperature variations), by jointly learning from time series corrosion data collected from corrosion sensors installed in Gangseo-gu, South Korea to predict the corrosion rate accurately. The CGRU model has been reported to outperform established models—such as CNN, GRU, LSTM, and DNN—in terms of accuracy, convergence rate, and robustness against environmental noise. Furthermore, the model was shown to be interpretable and sufficiently practical by sensitivity analysis, showing that solution resistance (salt concentration) is the most important factor contributing to corrosion. The proposed framework is a powerful tool for the integration of digital twins and data-driven maintenance strategies for offshore infrastructure and provides direct implications on improving safety and decreasing lifecycle cost. In parallel, (Yang et al. 2024) proposed a multi-stage prediction framework for the estimation of corrosion damage as well as the

remaining life of oil and gas pipeline structures. To predict the corrosion rate with very high accuracy in the range of 0.0023 and 0.17% RMSE and MAE, respectively, they adopted the Moth –Flame Optimization (MFO) algorithm with the Least Squares Support Vector Machine (LSSVM) to develop the MFO – LSSVM model. Based on this, they also employed a hybrid LSTM- DNN model to predict the pipelines' remaining useful life, which could predict the inherent time-dependent and nonlinear characteristics of the pipelines with MSE = 0.0004 and R2 = 0.9999 (Cui et al. 2024).

A robust digital twin framework for pipeline condition monitoring was proposed by (Wang et al. 2024). This model integrates real-time sensor data, probabilistic modelling, and augmented reality to provide a dynamic and high-fidelity virtual replica of the structure of physical pipeline systems. This framework uses physics-based simulations, Bayesian inference, and data-driven surrogate models for identifying and predicting the specific area and time of the structural degradation (such as fatigue cracking at a welded joint). The novelty of this method is the combination of physical and virtual systems, which is a milestone in the intelligent structural management of industrial pipeline networks.

Integrated predictive systems are highly important for slope-induced pipeline failures, which, despite the use of state-of-the-art practices, are often driven by ground instability and associated geometric imperfections. Longitudinal compression, insufficient friction and overbending can lead to buckling of the pipes in steep terrain, resulting in catastrophic failures. While conventional studies explain these failures with the help of mechanical models (Palmer et al. 1999), there are no tools for foreseeing such events in dynamic scenarios. Incorporating the information on the material deterioration into collapse predictions is reported in (Lignos and Krawinkler 2011), which helps in improving the seismic endurance of steel frames. Models that simulate the deterioration of steel components underloading were similar to those experienced in earthquakes. The work helped to assess the safety of steel frame structures not only by considering their initial strength but also by evaluating their performance under dynamic conditions, such as seismic loading. Gaps identified are limited consideration of cyclic loads, insufficient representation of component-level behavior and lack of framework for unified collapse prediction. Figure 4 presents the interrelation between corrosion and structural instability in engineering systems, highlighting how AI models can predict degradation under complex loading.

Precious studies broadly trend toward active rather than passive asset management of structural systems. Through embedding AI within traditional 'deterioration' models, engineers can no longer need to rely on deterministic failure thresholds but can develop continuously and fractionally monitored risk-aware systems. An obvious way to increase the research impact will be to couple physical models with an explainable AI framework for transparent and traceable

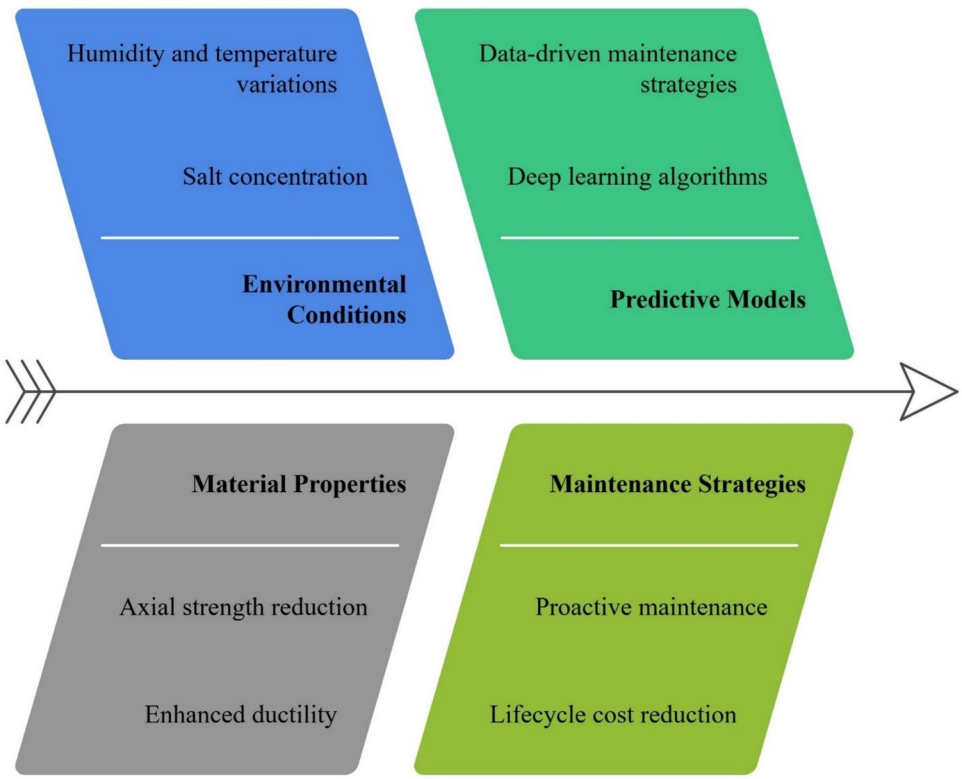


Fig. 4 Addressing corrosion and structural instability in pipelines

structural failure prediction in pipelines. The use of multi-source data from the structural health monitoring sensors, visual inspection imagery, and geotechnical indicators can improve the prediction accuracy. Additionally, implementing digital twins across various domains supports the development of interoperable platforms for real-time decision-making and lifecycle management. This approach facilitates sustainable and data-centric models, leading to enhanced safety and integrity of the pipeline structures.

2.4 Cluster 3 failure analysis

Traditionally, asset integrity management on pipelines has relied on empirical models, periodic inspections and 'traditional' failure analysis methods for evaluating pipeline reliability. However, most of these approaches are limited in their failure prediction accuracy, especially in dynamic operating environments where corrosion, mechanical and structural defects and pressure fluctuations interact together with complexity. In recent years, different aspects of failure assessments have been studied through research efforts, which explains how AI can change the conventional procedures in maintenance strategies.

For example, predicting internal corrosion and resulting failures of crude oil and natural gas pipelines using machine learning is reported in (Fang et al. 2023). Currently, the existing corrosion prediction models are based on empirical relationships and historical failure data, and they do not explain the complexity of non-linear and multi-factorial progression of corrosion leading to failures.

Although corrosion is a primary mode of failure, mechanical defects and structural weakness contribute relatively large fractions to the integrity of a pipeline. This can be assessed through understanding the failure pressure in high-strength pipelines with corrosion defects, as shown in (Ma et al. 2013). Using numerical simulations and experimental validation, this study shows how corrosion weakens mechanical behavior, which can cause failure. Using real-time defect growth monitoring or monitoring of historical effects shows that failure pressure predictions can be far more accurate than the static assessments. By continuous learning with new data, machine learning may augment this approach to update the failure pressure estimates as corrosion pattern and the environment change over time.

The effect of defect shape and orientation on the burst pressure of pressurized pipelines has been further investigated in (Al-Owaisi et al. 2018). Experimental results proved that the burst pressure varies significantly because of the complexity of the structural failure mechanisms resulting in different defect geometries. The variability of such defect type presents a major challenge in conventional failure analysis methods, which generally assume simplified defect shape. However, AI-driven models can be generalized among the disparate defect geometries to reveal failure trends that are not obvious through other means. With training of deep learning models on very large datasets of actual defect cases, better estimations of burst pressure can be achieved through more accurate predictive frameworks for pipeline safety assessment.

Studies are also conducted on the influence of the yield-to-tensile (Y/T) strength ratio on the burst pressure of corroded high-strength pipelines (Chen et al. 2017). Results of this sturdy contributed to an understanding of how material properties influence failure in pipelines that are usually neglected in generic failure models. Moreover, incorporating material property variations in the development of AI-based predictive models would lead to better insights in failure risk assessments in various pipeline compositions as well as the manufacturing process. A solution based on the hybridization of physics-based simulations and pattern recognition, enabled using AI, can improve the burst pressure estimation. Figure 5 illustrates how incorporating AI into the process enhances the understanding of complex failure mechanisms in pipelines and improves the accuracy of failure predictions.

Using AI for a comprehensive assessment of energy pipeline degradation is presented in (Hussain et al. 2024). Challenges in collecting high quality data, integrating predictive analytics with current asset management systems, and reliable use of AI models in real-world applications are outlined in this study. It is noted that treating AI models as black boxes—without explaining their predictions—raises significant concerns among industry stakeholders, who require interpretable and justifiable decision-making processes to ensure regulatory compliance and maintain operational trust. This is addressed in the study by developing explainable AI (XAI) frameworks, which could significantly improve the usage of AI-based solutions in the industries.

In spite of these studies, several research gaps still exist in this area, which are yet to be addressed. For example, most of the current AI models mainly center around static data analysis. A more dynamic approach, which integrates the real-time sensor data, along with real-time environmental conditions and operational history, can improve the failure prediction capabilities. Furthermore, the AI models need to be constantly trained with new failure datasets to improve its accuracy over time.

While many existing studies on wear are based on historical failure data, the emergence of new Industrial IoT (IIoT) technologies, particularly advanced wear sensors, now enables real-time, high-resolution monitoring. Incorporating IIoT-enabled AI systems which uses distributed sensor networks and adaptive learning algorithms for real-time pipeline health monitoring is a promising direction for future research. Further, hybrid AI models could fuse deep learning with physics-based simulations, which can improve the model performance significantly. Physics based models

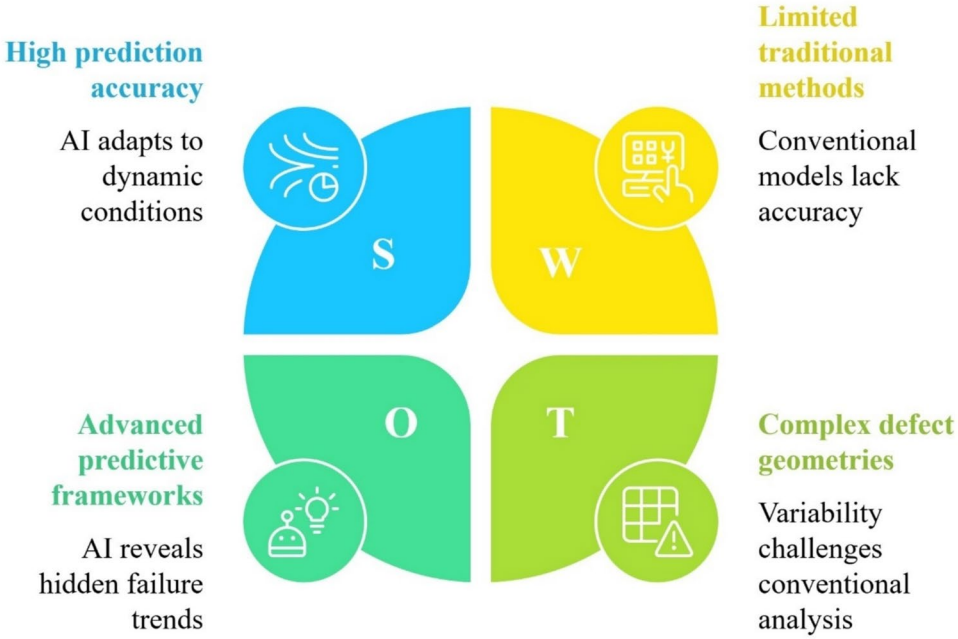


Fig. 5 AI-driven pipeline integrity management

help in getting a fundamental understanding of pipeline behavior under varying loads which can then blended with AI to understand complex failure patterns. This could result in a more robust predictive maintenance framework that is adaptive to different pipeline conditions with interpretations.

Current AI models are typically evolved from a single type of dataset e.g., corrosion measurement or failure pressure estimation. Nevertheless, pipeline failure can be regarded as a combination of mechanical, environmental and chemical factors. AI-based future systems should use various relevant information, for example, corrosion rates, pipeline material properties, operating pressure, environmental conditions, and real-time sensor readings. With this approach, these pipelines can be assessed more accurately and holistically, which can eliminate the risk of unexpected failures.

With industries moving toward digital transformation, predictive maintenance based on AI can help in assuring the reliability and sustainability of critical infrastructure.

2.5 Cluster 4 adaptive integrity assessment and management

In recent years, pipeline integrity management has begun to evolve from conservative, rule-based approach toward more dynamic, data-informed strategies. This shift is particularly evident in research that combines advanced risk-based models and uncertainty analysis to derive intelligent and adaptive integrity management.

A fundamental limitation of the structural integrity models for pipelines is that they assume highly specific conditions under which the plastic collapse of the pipeline occurs. For example, conventionally used criterion like the Tresca and von Mises are inadequate to determine burst pressure for modern high strength pipeline steels. As this drawback of the classical models exists, (Zhu and Leis 2006) has provided the Average Shear Stress Yield (ASSY) criterion, which operates between these two classical models to offer a more balanced assessment of the multiaxial stress states. It is shown that neither the lower bound Tresca nor the upper bound von Mises criterion is applicable to modern pipelines. Instead, an ASSY criterion is developed that is compared with experimental burst pressure data and provides a more representative prediction of the burst pressure for isotropic hardening materials.

While this development is mathematically sound and experimentally validated, it does not consider the offshore and subsea operating environments, which are characterized by conditions such as cyclic loading. This information could be better captured through real-time integration of sensor data. Figure 6 shows the possible evolution of pipeline integrity management to be more adaptive approach.

Another key factor influencing the mismatch with real-world variability is corrosion allowance design. Historically, design codes posit a constant corrosion allowance (typically between 3 and 6 mm) regardless of the environment the system will operate in or the expected degradation rate. An approach in (Hasan et al. 2018) suggests a risk-based corrosion allowance framework, which is more realistic and adaptive to the real operating conditions. Instead of assuming uniform corrosion behavior across all installations, their approach considers the actual operational risks arising from environmental exposure to calibrate protection strategies more effectively. This a risk adaptive methodology is

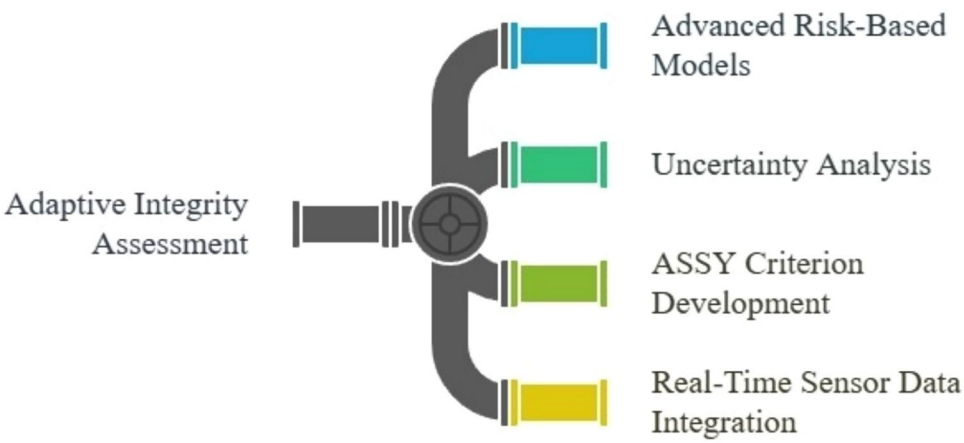


Fig. 6 Evolving pipeline integrity management strategies

particularly useful for extending service life while minimizing the over design. However, such approach only partially covers for unpredictable acceleration of corrosion due to factors such as seabed shifting, marine biology interactions or dynamic mechanical loads, which are complex external influences.

A better understanding on this issue, particularly in estimating the failure likelihood of pipelines, can be brought through probabilistic modeling of pipeline degradation. A probabilistic transformation method (PTM), with a gamma distribution model for calculating the failure probability regarding individual corrosion defects and the left-over risk of pipeline segments post repair, is proposed in (Younsi et al. 2013). Unlike the Gaussian models, gamma gamma-based approach could make more reasonable prediction of failure in short intervals following inspection, which can avoid unnecessary repairs. At the core, their method is based on actual inspection data from smart pigging runs and takes into account the stochastic corrosion growth as well as the uncertainty in the resistance parameters. This work stands apart in being both practical, as its framework can directly be used for development of maintenance scheduling and rehabilitation planning under situations where highly conservative models tend to lead to costly schemes. These together comprise a coherent approach toward context-aware, risk-informed and probabilistic based pipeline integrity strategies. Pipeline systems should not be considered as a fixed parameter system, rather, they are to be considered as a dynamic probabilistic system.

The integration of AI would enhance the ability of such models to adapt to stress, corrosion, and environmental fluctuation patterns that are not easily handled with physical models. Future work needs to be focused on integrating sensor-based monitoring systems with AI enhanced predictive models to make the pipeline integrity management adaptive. By adding the machine learning algorithms trained with the live operational data on simulations of physics-informed models such as ASSY, predictions regarding the yield points, burst pressures and the localized failures could be improved. Furthermore, through digital twin platforms (or virtual representations that evolve concurrently with physical counterparts), one can simulate real-time stress and corrosion state, which can help operators in anticipating failures and there by optimizing maintenance. When integrated with marine environment information such as seabed dynamics and biological fouling, the management strategies can be made holistic. Figure 7 details the proposed risk-based corrosion allowance cycle, which highlights the shift from the fixed corrosion allowances to an adaptive approach based on data driven factors.

2.6 Cluster 5 probabilistic and stress-based modelling

Complex loading environments and long-term corrosion have become more challenging to the structural integrity of pipelines in energy infrastructure. As explained under Cluster 4, studies with classical deterministic models are not able to describe all sources of uncertainties involved in pipeline operation, particularly when cyclic loading and progressive corrosion are involved. Hence, there is a research shift towards more probabilistic, time-dependent, and simulation-driven approaches for pipeline failure prediction and risk management.

A significant contribution in this area is reported in (Qin and Cheng 2020) in which the combined effects of corrosion defects and cyclic mechanical loading (induced during in-line inspection operations) on pipeline failure behavior is analysed using 3D finite element modelling (FEM). Their model considers critical parameters such as defect depth, internal pressure, and stress ratios (R-ratios), revealing that cyclic loading significantly increases localized stress and strain at defect sites. This results in a notable reduction in failure pressure, which is not captured by traditional burst models. What makes this work particularly valuable is its ability to simulate stress/strain

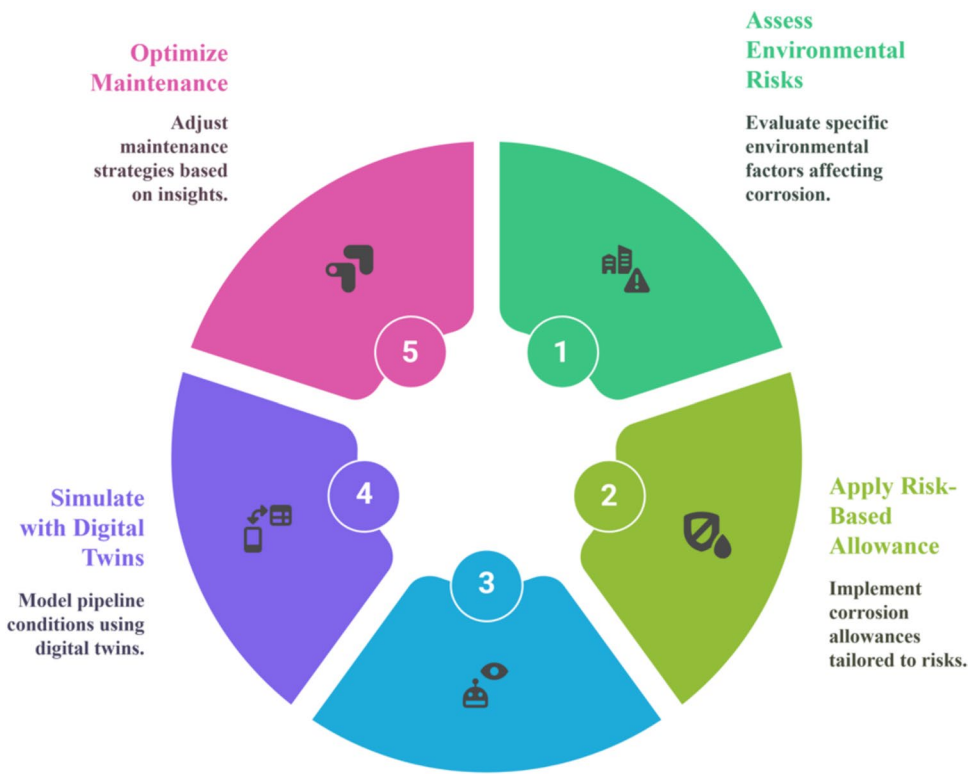


Fig. 7 Risk-based corrosion allowance cycle

concentration under realistic operational disturbances. However, while the model is validated against experimental data, it still lacks real-time adaptability or integration with live sensor feedbacks.

Complementing this, a new failure pressure prediction model based on FEM, combined with Monte Carlo simulation to conduct a full probabilistic reliability analysis is proposed in [33] Their model improves upon conventional burst pressure formulas by introducing geometry-sensitive variables and a sensitivity analysis framework. The study identified corrosion depth and internal pressure as the most influential variables affecting failure probability. This marks an important step forward in risk quantification. However, the model remains static and does not use real-time data ingestion or pattern recognition capabilities that AI-based systems could offer.

Zhou (2010) introduced a stochastic system reliability framework, accounting for multiple active corrosion defects and varying internal pressure loads over time. The study models the pipeline as a series system with distinct failure modes (small leak, large leak, rupture) and shows how spatial variability in pressure and corrosion growth significantly affects system reliability, more than the variability in pipeline material properties. This work contributes to the analysis by considering interacting defects and system-level reliability to the analysis. However, it remains as a simulation tool and cannot be used as a predictive or autonomous decision-making system.

Studies discussed above emphasize the critical importance of modeling corrosion growth under realistic operating conditions, especially under cyclic fatigue and spatially varying environments. However, possibilities of integrating AI for further improvements are not considered in these studies. While probabilistic FEM methods offer accuracy, they do not support real-time learning, scalability across varying pipeline contexts, or automation in detection and decision-making. To advance the current state of pipeline integrity assessment, future research should focus on the integration of AI and machine learning by incorporating deep learning models—such as LSTM and CNN-LSTM hybrids—with both historical and real-time inspection data. This would enable the system to learn degradation patterns under cyclic loading and fluctuating operational conditions. Simultaneously, the development of real-time digital twin systems tailored to pipeline infrastructure can allow for the continuous simulation of failure progression, real-time updates, and early warning alerts. By combining physics-based simulations with AI, these digital twins can significantly enhance predictive accuracy. Furthermore, implementing multi-scale modeling that fuses data from high-resolution in-line inspections (ILI), SCADA systems, and acoustic or vibration sensors will provide a more comprehensive assessment of localized stress concentrations, fatigue-prone zones, and corrosion hotspots.

Finally, integrating these predictive tools with risk-based maintenance decision frameworks can enable the automation

of inspection and repair scheduling, ultimately enhancing cost-efficiency, safety, and responsiveness in pipeline asset management. Figure 8 illustrates the advancements in pipeline integrity assessment, highlighting the integration of the proposed future developments in managing the integrity of oil and gas pipeline systems.

2.7 Cluster 6 pitting corrosion

Oil and gas pipelines are highly sensitive to the pitting corrosion as it can lead to serious structural integrity issues, potentially causing leaks or even catastrophic failures. The study of pitting corrosion in pipelines relies on statistical models because this phenomenon follows random distribution patterns as discussed in Click or tap here to enter text. (Caleyo et al. 2009; Sheikh et al. 1990). Application of extreme value statistics to analyze maximum pit depths and leak timing began through the pioneering research in (Sheikh et al. 1990) which revealed the Weibull extreme-value distribution as the distribution that best represents time-to-first-leak data. Monte Carlo simulations, integrated with pit growth mechanisms, revealed that the pipeline age affects the evolution of pit depth distributions (Caleyo et al. 2009). Short-term corrosion data can match Weibull and Gumbel statistical models until pipe exposure reaches long periods that produce Fréchet distributions through diffusion-limited pit growth behavior (Caleyo et al. 2009).

These statistical investigations are further advanced through simulation techniques that quantify pipeline failure probability under corrosion. Time dependent reliability models (both for burst pressure failure and stress-based failure) were developed and presented in (Liu et al. 2019) which indicates that Monte Carlo simulation (MCS) could be augmented with more efficient methods. They also emphasize the importance of including multiple loading conditions (soil pressure, residual stress, thermal expansion, bending) that have effects on failure probability (Liu et al. 2019). However, during the pipeline service life, the corrosion rate parameters (i.e., depth growth coefficients) are the main factors which contribute to the reliability degradation. Simulation efficiency has been further improved during recent research (Ben Seghier et al. 2022) by introducing a weighted depth difference method to a subset simulation approach. This could address multiple interacting corrosion pits with an ability to make accurate failure probability estimates for pipeline with irregular corrosion defect profiles. Comparative studies were carried out on different simulation and surrogate modeling strategies, which showed that data-driven MCS can give high accuracies even under reduce computational demands. An artificial neural network (ANN) surrogate, when coupled with Monte Carlo Simulation (MCS), was able to reproduce the failure probabilities with approximately 98% accuracy compared to direct simulations, at the same time require only much lower computational demand (Ben Seghier et al. 2022). Their combination improves the predictive reliability analysis of corroded pipelines, demonstrating how Monte Carlo, subset simulation and the AI-based meta models can be used to enhance the analysis.

Empirical models such as ANSI B31G (American National Standards Institute) have historic application as industry standards to estimate capacity of corroded pipelines while the industry has progressed to more advanced, physics-based methods. The burst pressure is estimated based on idealized flaw shapes and B31G and RSTRENG correcting

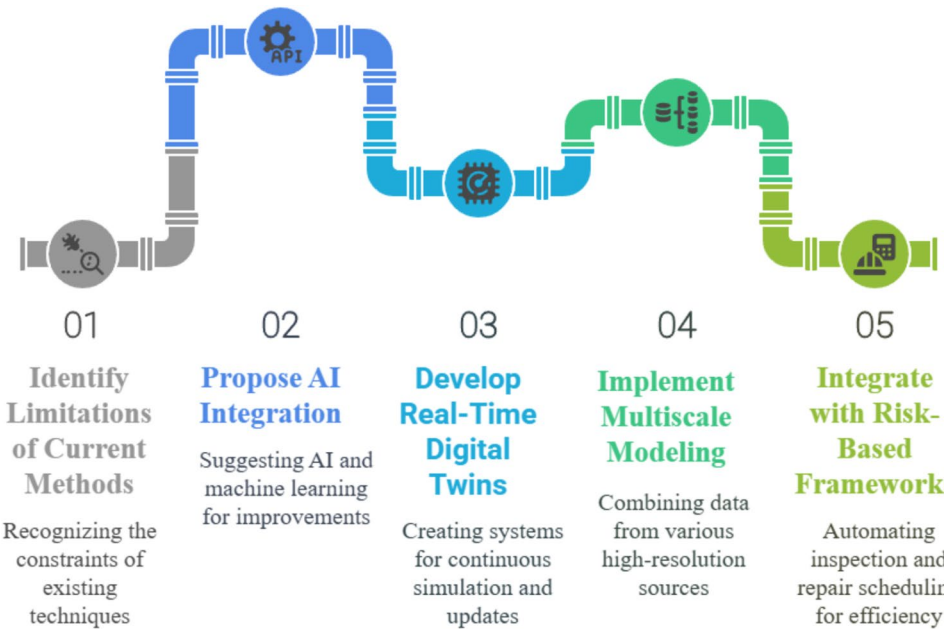


Fig. 8 Enhancing pipeline integrity assessment

error respectively by accounting for irregularity in pit geometries. Other tools such as the PCORR (Pipeline Corrosion and Risk Reduction) program had further increased the level of precision by performing a finite element analysis, which takes into account complex stress conditions (Wilkowski et al. 2000). The burst tests provide the validation to these models which are now used as the basis of fitness-for-service evaluations.

In recent days, application of artificial intelligence (AI) and IoT-based monitoring are being utilized in pitting corrosion management. With these applications, we can now identify and detect the size corrosion pits automatically from inspection and sensor data (Faraji 2025.). Digital twin technology serves as a real time monitoring tool in the form of virtual models that remain updated with real-time field data to reflect the real pipeline condition (Chen et al. 2025). These high-fidelity digital twins with AI can be used to predict pitting corrosion and notify the defects ahead of time with over 90% accuracy. (Faraji 2025). These hybrid AI models take a proactive and adaptive pipeline integrity management approach by forming a bridge between mechanistic domain knowledge and data-driven insights. Validation and standardization of these advanced methods are progressing. The above discussed advancements in pitting corrosion management are illustrated in Fig. 9.

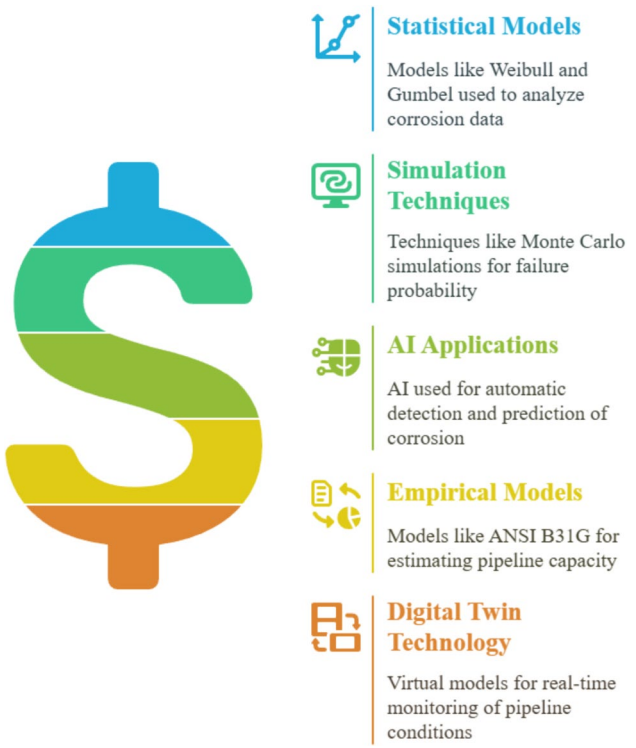


Fig. 9 Advancement in pipeline corrosion management

2.8 Cluster 7 dented pipelines

External mechanical actions cause dents in pipelines, which, under internal pressure, are a significant concern for integrity. These deformations may cause local stress concentrations and degrade pipeline strength and fatigue life.

According to a recent study (Karamanos and Andreadakis 2006; Iflefel et al. 2005), the dent behavior of pressurized pipes was analyzed using finite element modelling and along with simplified analytical approaches. In this study, dent simulations were performed under the design pressure, and the dents were then tested under bending. Variation of internal pressure had influences on the dent performance, mostly beneficial for the tension-side of the dents. However, this can adversely affect the compression-sided dents. These findings are very important for assessing the structural integrity, as they consider both the dent location and pressure levels.

A nonlinear FEM framework, capable of accurately reproducing experimental load displacement responses of dented pipes, is presented in (Karamanos and Andreadakis 2006). Based on this model and 3D analytical tools for plastic deformation, pressure effects, boundary conditions and yield anisotropy related to dented pipelines are presented. It was found that an increase in internal pressure increases denting resistance and changes the deformation geometry. The dent behavior was found to be highly influenced by anisotropy in yield strength (common in manufactured pipes), especially under axial constraints.

Finite element methods (FEM) are proposed to incorporate residual stresses, anisotropy, and multi-defect interactions on dented pipes (Karamanos and Andreadakis 2006). Additionally, multiscale sub-modeling strategies are proposed which integrate precalculated dent analysis into full pipeline simulations, enhancing accuracy while reducing computational costs. This could also integrate with digital twin systems for real-time structural monitoring.

It was shown that when the datasets from simulations and experiments are provided to the deep neural network, the latter predicts the burst pressures of dented X-grade pipelines with high accuracy (Oh et al. 2020). Similarly, a report by (He and Zhou 2024) also demonstrated that with generative models which augment dent gouge data, the prediction capability of the model can be improved far beyond the traditional empirical models. In-line inspection analytics are also being used with AI applications to identify high risk dents as well as to prioritize its management. The models developed under these studies are further to be validated through full scale testing. Further, recent studies indicate that the current 6% strain-based limits of depth and strain (and others) may be highly conservative (Abdelmoety 2023). The allowable strains can reach more than 20% in some cases, which indicates the need for strain-based assessment which account for the dent curvature and localization in failure analysis.

Future research should aim at integrating AI with high-fidelity simulations through digital twins, which are real-time virtual models that combine sensor data with AI to predict dent evolution. The failure criteria should include several parameters like dent geometry, material properties, and pressure history. Expanding experimental validation will help to improve the accuracy of these models which can be used in the predictive maintenance of oil and gas pipelines. Figure 10 shows the various pipeline dent assessment methods discussed above, comparing traditional methods of finite element modeling with AI-assisted prediction tools.

2.9 Cluster 8 pipeline system integrity

With recent progress in pipeline integrity research, models that integrate internal corrosion and material dependent reliability into probabilistic assessment are being developed. The impact of uncertainty in operational conditions and material behavior, however, cannot be incorporated into traditional deterministic models, particularly for high-strength pipelines. A study (Maes et al. 2008) presents a detailed reliability framework that incorporates variations in the yield-to-tensile strength ratio (Y/T), demonstrating how high-strength steels such as X100 and X120, which has lower strain hardening, respond differently to corrosion damage compared to standard steels with higher strain hardening. The study also finds that, for steels with higher Y/T ratios, higher internal pressure, bending, and axial loads can be incorporated into their multiaxial finite strain approach. The results show that deeper corrosion grooves lead to a more rapid reduction in burst capacity. They incorporate constitutive behavior into structural reliability, highlighting the need to adjust design limits based on grade-specific properties and long-term degradation.

In complement to this, (Shabarchin and Tesfamariam 2016) suggests a Bayesian Belief Network (BBN) framework for the internal corrosion hazard assessment in oil and gas pipelines. Their model includes pitting, erosion, MIC, pipe geometry, inhibitor effectiveness, and fluid chemistry integrated with a general power law model. Using field data, expert judgment, and analytical models, the BBN estimates probabilistic failure likelihood for each pipeline segment. This approach is capable of dealing with uncertainties and sparse data, which is crucial for the reliability of aged pipe infrastructures under complex operation scenarios.

Use of machine learning for continuously updating the corrosion growth rates based on sensor inputs is reported in (Hu 2024). Similarly, (Ismail et al. 2011) proposed a hybrid AI-Bayesian models for better prediction of failure probability under varying environmental conditions. For future systems with digital twins and predictive maintenance, integration of simulations and probabilistic reasonings are increasingly essential. Possible AI-driven pipeline integrity has various steps, starting from data acquisition to predictive maintenance, as shown in Fig. 11.

The ongoing review and analysis underscore the evident need and importance of applying AI in Asset Integrity Management. Future research directions for each specific cluster area have also been outlined in the respective sections. Implementing these recommendations is expected to enhance and streamline integrity management for oil and gas pipelines, offering significant advantages in cost efficiency, time management, and reduced man-hours for inspection and monitoring. This, in turn, contributes to improved process safety management.

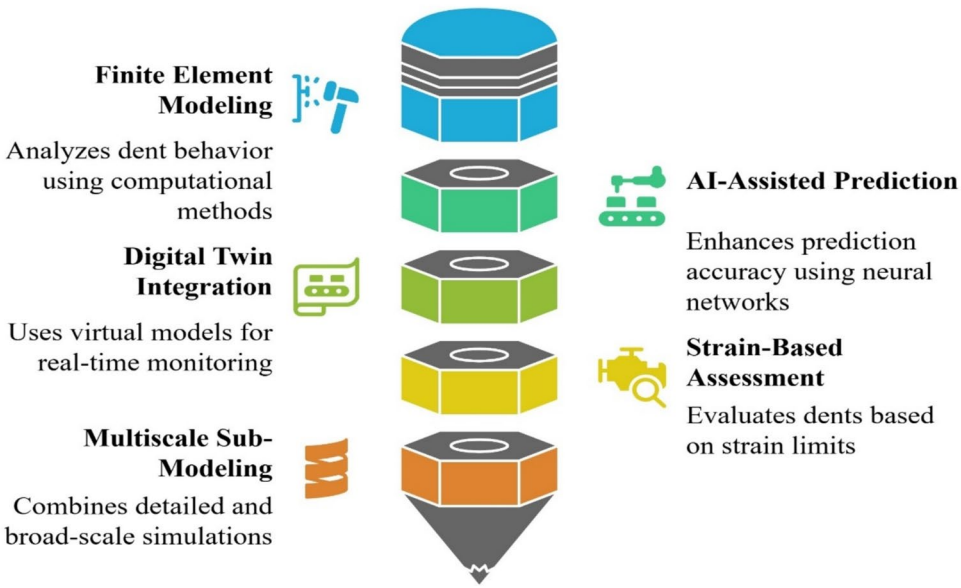


Fig. 10 Evaluating pipeline dent assessment methods

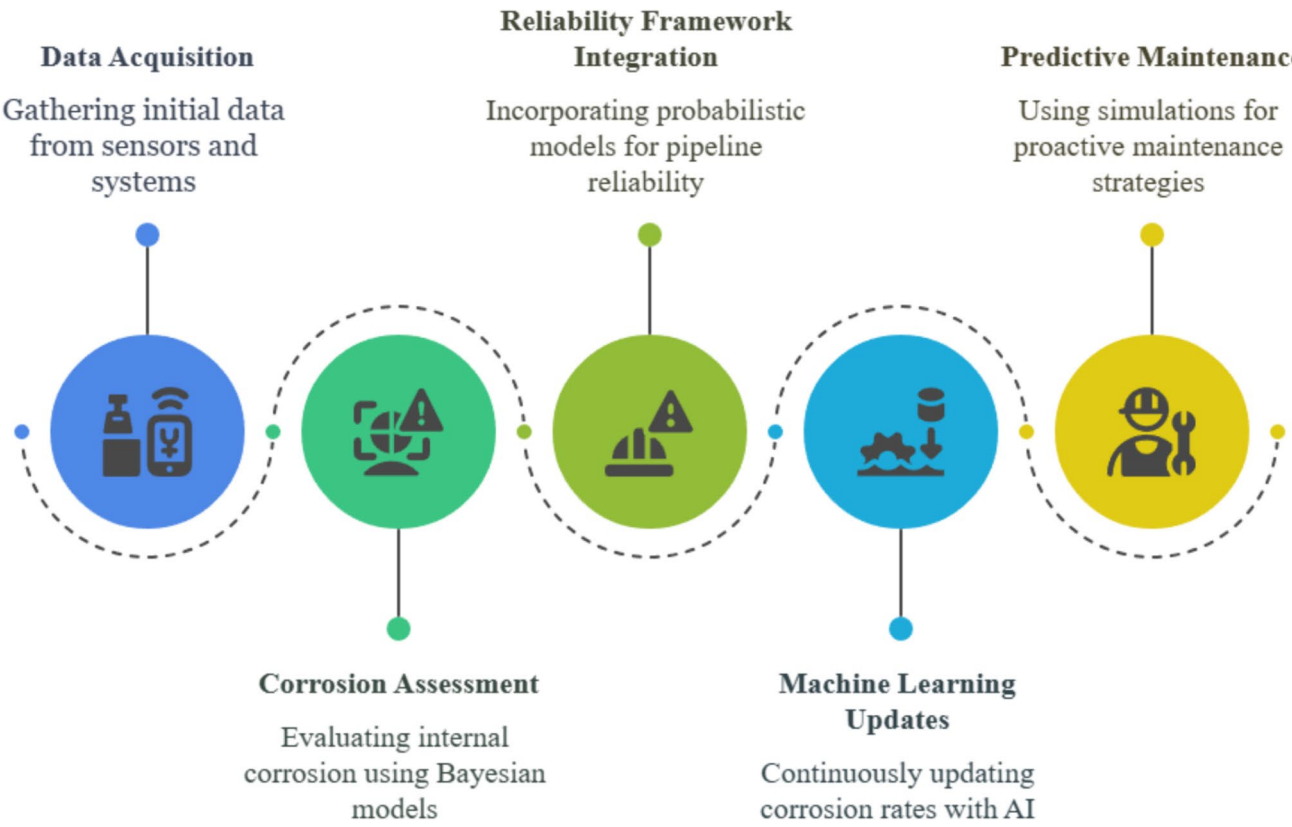


Fig. 11 AI-driven pipeline integrity framework

3 Artificial intelligence (AI)-based asset integrity management (AIM) system

Based on the above analysis, an AI-based asset management system is proposed for pressure equipment consisting of piping and pressure vessel assets in the offshore oil and gas platforms. The system integrates both qualitative and quantitative methods, offering a comprehensive approach that uses real-time data analytics, machine learning, and predictive modeling. By enabling a more adaptive and efficient asset management strategy, it addresses the limitations of existing standalone methodologies followed and reduces reliance on traditional empirical practices. A generic offshore oil and gas asset which has already surpassed the operational life is considered under this analysis to capture the typical challenges faced by such aging facilities worldwide. The objective is to reduce operational costs, minimize human exposure to hazards, and extend the platform's useful life through proactive maintenance and repair strategies.

Figure 12 illustrates the proposed AI-based AIM Framework which consists of the 3 key components.

3.1 The proposed three-phase asset integrity management program

The Oil and gas industry has relied on well-established AIM Frameworks based on established industrial standards such as API 510,570,580, and 581, which provide structured approaches for inspection planning, degradation management, and compliance.

The proposed AI-based AIM Framework builds upon complementing these traditional standards. While API based procedures operate on fixed intervals and are largely deterministic by taking input from inspection intervals and risk matrices, the proposed system introduces real-time sensing, data-driven prediction, and AI-enhanced decision support to proactively manage integrity.

The key differentiators are:

- Real-time monitoring and predictive analysis, in contrast to periodic inspection schedules
- Dynamic risk profiling using AI and Machine learning tools (e.g., CNNs, LSTM, Bayesian models) instead of static risk categories.

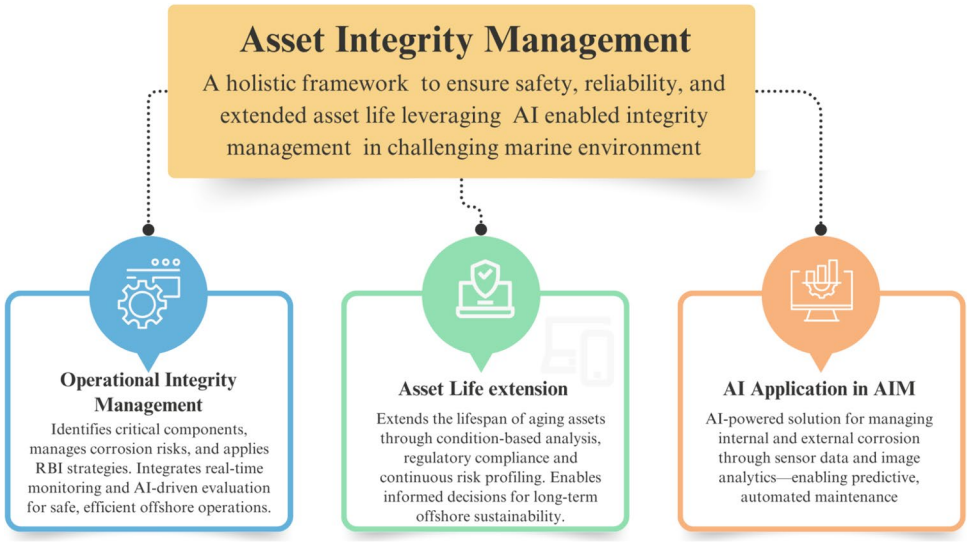


Fig. 12 Proposed AI-based AIM framework

- Explainable AI overlays (e.g., SHAP, Grad-CAM) that provide transparency and auditability aligned with compliance expectations.

These enhancements are not aimed at replacing the established standards but rather elevating them with continuous monitoring and automation, offering condition-based monitoring and risk-informed, cost-effective decision-making in maintaining integrity.

The proposed method aims to enhance asset integrity management by integrating traditional practices with modern AI technologies through a structured, three-phase Asset Integrity Management (AIM) program, as illustrated in Fig. 13. The first phase, Operational Asset Integrity Management, serves as a foundation focused on defining a framework for managing the integrity of the platform-an essential prerequisite for all subsequent analysis and interventions.

Building on this, the second phase addresses Asset Life Extension by targeting the long-term sustainability of the asset. This phase emphasizes extending operational life by monitoring and managing critical parameters specific to the health of aging assets. The final phase introduces AI Applications in Integrity Management, where artificial intelligence tools are employed to optimize the procedures established in the earlier phases. This integration of AI enables automation, enhances predictive capabilities, and supports improved decision-making, ultimately leading to increased operational efficiency and reliability.

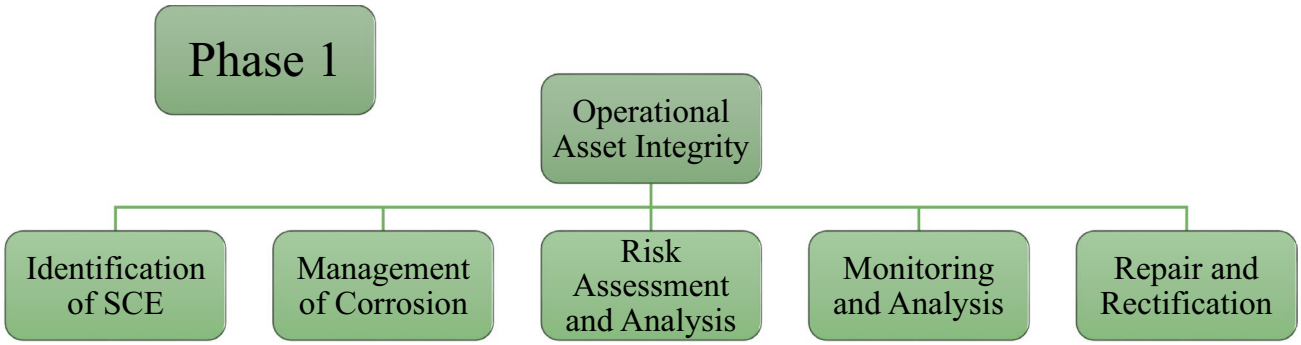
The three phases are interconnected, with outputs from one stage driving another for smooth information exchange and ongoing enhancements in asset management practices. With this approach, the day-to-day operations at the platform can be performed better, resulting in safer operations and longer life under challenging offshore environments. This global strategy aims at preserving asset integrity throughout its life cycle until it is decommissioned. This subsequent section discusses the details of each phase.

3.1.1 Phase 1—operational asset integrity management system

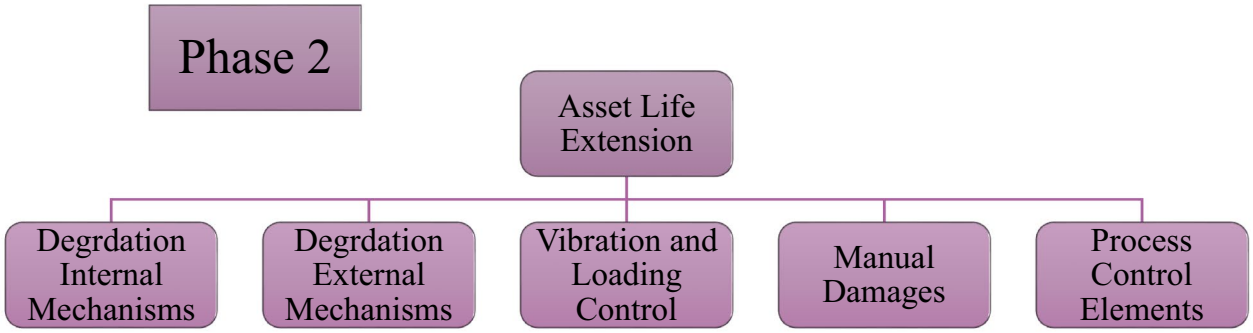
The first phase of the Asset Integrity Management program details the fundamental elements that need to be assessed, defined, and controlled to ensure the operational safety and functionality of the asset. This phase elaborates on how the 5 major identified elements are defined and assessed through a systematic approach.

3.1.1.1 Identification of safety critical elements (SCE) This phase identifies and manages Safety Critical Elements (SCEs), whose failure can lead to major accidents. Asset risk and criticality are assessed using HSSE Case inputs to identify SCEs. This approach helps to perform risk ranking and prioritizes SCE monitoring and maintenance.

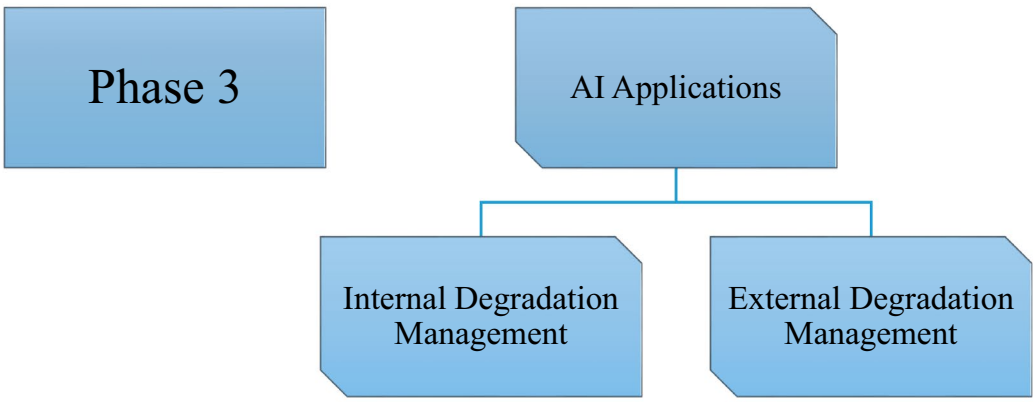
Figure 14 shows the procedures, characterizing the identification and management of the SCEs. The first process of risk assessment is known as risk appraisal and is defined as the identification of hazards that have the potential to threaten the integrity of the asset. This is succeeded by grouping and barrier definition, where hazards are recognized and countermeasures are developed to deal with any threats that may have been identified, thereby forming the risk management process. A detailed register of all relevant SCEs is then created, which lists all the prospective factors and the potential barriers linked to each one of them. Subsequently, steps for operation are created. Finally, for constant and efficient SCE information monitoring, all information is registered in the system.



(a)



(b)



(c)

Fig. 13 Three phases of asset integrity management (AIM); **a** phase 1, **b** phase 2, **c** phase 3



Fig. 14 SCE identification-workflow

3.1.1.2 Management of corrosion elements Operating under severe marine corrosive conditions, offshore platforms rely heavily on corrosion control to maintain its structural integrity (Wood et al. 2013). Corrosion management of the platform consists of regular checks and proactive control tools (corrosion inhibitors, cathodic protection, corrosion-resistant materials, and corrosion-resistant coating systems).

Objective of the Corrosion Management Framework (CMF) of the platform is to provide a holistic approach in managing internal and external forms of corrosion that exist due to the marine nature, operating conditions, and process medium. At the design stage, materials resistant to corrosion are selected, and these corrosion control parameters are maintained throughout the operation phase. The three sections such as Threat Identification and Categorization; Controls and Mitigations; Monitoring and Reporting forms the framework of the CMF.

3.1.1.3 Risk assessment and analysis Risk assessment and analysis are vital to understanding the operational dangers and hazards of the platform. During this phase, traditional time-based inspection intervals are replaced with a risk-based approach to enhance the effectiveness of inspection plans. First, all the risks linked to each pressure component are critically evaluated, using the relationship

$$R_i = P_i \times C_i \tag{1}$$

where *Ri* represents the overall risk, *Pi* is the probability of failure, and *Ci* is the consequence of the component's failure. The risk-based inspection (RBI) program is based on (Festen 2024) and (American Petroleum Institution 2016).

This prioritizes maintenance and inspection tasks and further aids in having a systemic approach toward designing a high-level risk management mitigation strategy. Integrating such a well-rounded risk profiling phase effectively improves the predictive capabilities and operational safety of the proposed method by maintaining all parts at their highest level of safety and reliability based on a risk-based approach.

3.1.1.4 Monitoring and assessment The monitoring and assessment part AIM program includes continuous tracking and evaluation of operating parameters and the asset's condition. The system uses some of the diagnostic tools and technologies, which range from ultrasonic thickness sensors to scanners, cameras, drones, or other monitoring devices installed on pipes and pressure equipment.

Under this approach, an offshore data gathering module receives the data from sensors and wirelessly transmits it to the Computerized Maintenance Management System (CMMS) onshore. The CMMS, which integrates artificial intelligence and machine learning capabilities, checks incoming data against certain acceptance criteria. This

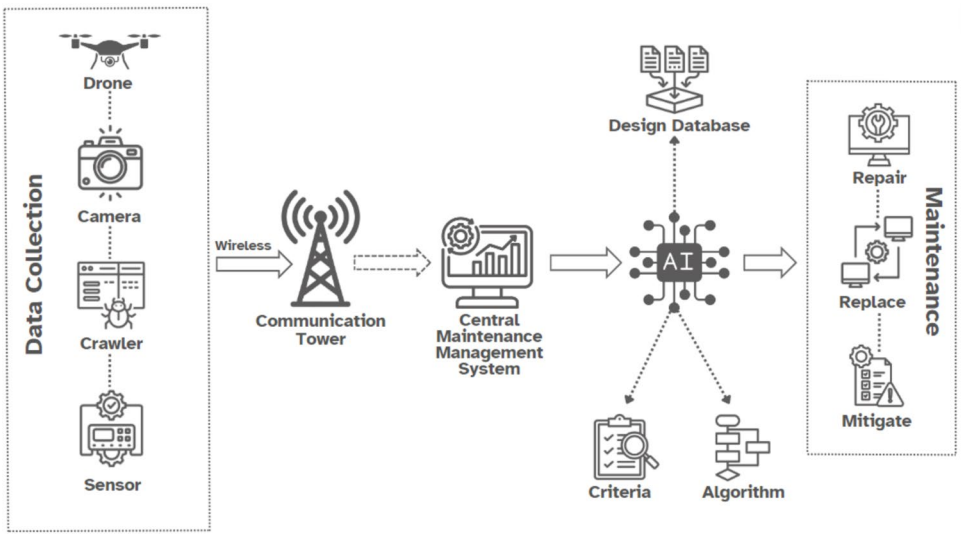


Fig. 15 Autonomous AI-based system and their workflow process

process aids in assessing the health of the asset, and the evaluation process (illustrated in Fig. 15) aids in precisely identifying the necessary maintenance actions.

AI powers the system with decision-making logic, which records sensor data, analyzes conditions, and triggers maintenance actions based on evaluation results. This end-to-end process with AI occurs automatically, reducing the need for human input and giving organizations a more proactive maintenance trigger—addressing potential issues before they become worse. The use of these advanced technologies in the AIM adds significant value to the asset management system's predictive capabilities with proper assurance of asset integrity.

3.1.1.5 Repair and rectification handling The key function of this phase is the repair and rectification, which is central in keeping the offshore platform operating in optimal condition. The goal of this stage is to rectify the deficiencies identified through continuous review and risk assessment. The equipment maintenance process is a mix of technical, administrative, and management activities throughout the project lifecycle to preserve the equipment's operational capability and perform functions as needed (Sattari et al. 2022).

Three types of asset maintenance regimes exist: corrective maintenance (CM), preventive maintenance (PM), and fabric maintenance (FM). The Computerized Maintenance Management System (CMMS) trigger corrective maintenance interventions when measurements fall below the specified threshold. To ensure continued safe operation, temporary repairs, such as synthetic material wrapping, and schedule permanent repairs are undertaken either within six months or during the next planned shutdown, depending on the severity and established criteria. However, preventive maintenance is planned and undertaken to quickly measure internal imperfections and make decisions based on risk-based inspection (RBI) studies.

3.1.2 Phase 2—asset life extension

This phase focuses on the strategies required to extend the operational life of the assets beyond their original design life. This phase is important as studies indicate that 70% of the offshore assets in Middle East are functioning beyond their normal design life of 25 years (Hovik 2015). To ensure their safe operation well beyond this design life, comprehensive management strategies are required.

Regulatory frameworks, such as the guidelines issued by the UK Health and Safety Executive (HSE), particularly the KP4 (UKHSE 2014) promoting asset life extension programs, play a crucial role in managing risks associated with aging infrastructure. These guidelines address age-related issues, staff turnover, process changes, and past inefficiencies in asset integrity management systems, with the goal of restoring and maintaining asset integrity beyond the original design life. Figure 16 illustrates the time progression of the asset's condition, highlighting critical milestones throughout its lifetime. Assessments should be conducted through a systematic analysis of the key influencing factors, identifying appropriate risk-mitigating actions necessary to manage all potential risks to the asset's integrity, ensuring safe and sustainable operations.

Asset Life Extension (ALE) barrier control plans are to be formed, especially for pressure equipment, to manage and

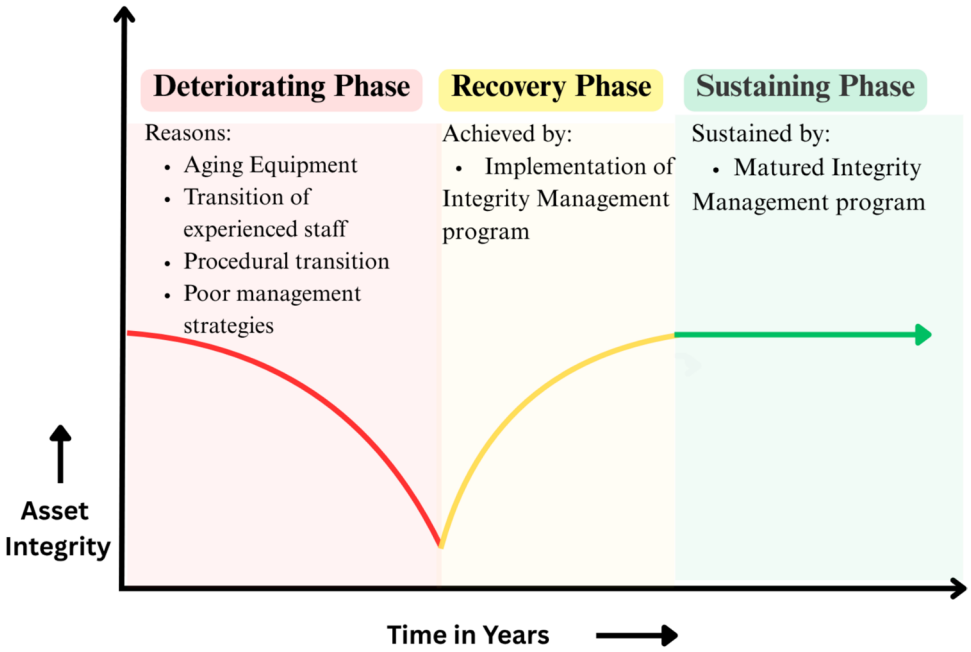


Fig. 16 Improving integrity performance

mitigate these identified risks effectively. The ALE Dashboard monitors these parameters identified and provides stakeholders with instant threat and health data. This dashboard provides insight into maintenance needs and ensures operational continuity by reporting key parameters and showing the performance of asset components. Dashboards provide insights into the current health of the system based on the risk factors identified for aging assets.

ALE is an important tool for offshore platforms that are operating beyond their design life and require an extended asset life. This phase proactively addresses technical challenges and vulnerabilities, such as corrosion, degradation, and equipment wear and tear, to help prevent accidents and safety incidents. This ensures that the platforms can sustain operational demands and uphold an optimal level of integrity over their extended period of operating life.

3.1.3 Phase 3—application

After implementing prospective improvements for managing the post-life integrity of the assets, this phase involves a shift from the basic computational and conventional methodologies to automated and digitized applications within the integrity management workflow. The proposed AI applications utilizes the advanced capabilities of machine learning to streamline integrity management tasks. This is achieved using automated data collection tools that interface with AI-integrated models, simplifying the process through real-time data inputs.

During this phase, AI applications are deployed in two key areas to automate the integrity and maintenance management of the platform, (1) Automation and Data Handling for Internal Degradation Integrity Management and (2) Fabric Maintenance for Effective External Corrosion Management.

3.1.3.1 Automation using machine learning for piping and pressure vessel internal degradation integrity management Majority of the piping and pressure vessels on the platform are made of carbon steel material, and few are of stainless steel. The most credible damage mechanism for internal corrosion is the CO2 corrosion present in the hydrocarbons and erosion by sand, produced from the wells. These are age related damage mechanisms and results in general and local metal loss. The prime factor to monitor to ensure the system health is the wall thickness of these components at the most susceptible locations called the thickness monitoring locations (TMLs). Installing ultrasonic sensors on the external surfaces makes it possible to measure the thickness of the wall continuously, without damaging it. This can be done in key places where pipes and pressure-retaining equipment are likely to be damaged over time by corrosion and erosion. A total of 98 TML locations could be selected where sensors can be placed to monitor the thickness.

In the proposed method, data are collected and input into an onshore central maintenance management system (CMMS) as in Fig. 17. Sensor data are processed and predictive modelling is deployed using a suitable algorithm like ensemble models such as random forest/XGboost to process the data and classify the anomaly category. Further maintenance scheduling is carried out based on the current health and predicted future corrosion failure. Reinforcement learning is proposed to optimize decision-making for maintenance scheduling based on the severity of the finding.

3.1.3.2 Fabric maintenance for optimum external corrosion management analysis Fabric maintenance plays a crucial role in reducing corrosion on external offshore assets, a significant contributor to containment incidents, accounting for more than 70% of these incidents. Fabric maintenance activities are a continuous requirement, so the process of recognizing, evaluating and rectifying are crucial to limit risks as well as extending asset life. According to (Spencer et al. 2019), AI-based approach can significantly improve the visual-based structural health monitoring.

Use of remotely operated vehicles (ROVs) to assess the condition of the coatings can provide detailed reports on asset coating conditions. Inspections yield different levels and states of corrosion, which are crucial to the analysis, as illustrated in Fig. 18. State-of-the-art software processes these pictures using image comparison methodology, matching them against a vast library of historical coating condition images.

This ranking is based on the mechanics of a standardized testing format to initiate appropriate maintenance actions based on the corrosion grade. The acceptance criteria should follow the RE grades in ISO 4628-3 Standard. Furthermore, fixed cameras located at key locations can continuously monitor the assets. The quality of the images is a critical factor in analyzing and comparing them to previous surveys. Advanced machine learning techniques can be used to analyze the images retrieved from these coating surveys. These models intelligently compare new images with predefined rust grading standards as well as past pictorial data to make an accurate evaluation of the fabric condition of an asset. The model then triggers appropriate fabric maintenance corrective actions from their respective maintenance databases based on these detailed rust grading comparisons.

The system is trained to identify the condition of paint and accurately evaluate corrosion patterns to ensure correct judgment on the maintenance steps that are required. Figure 19 illustrates a typical logic built to identify the appropriate maintenance action based on the rust grade identified. The stakeholders receive periodic updates via dashboard alerts and emails.

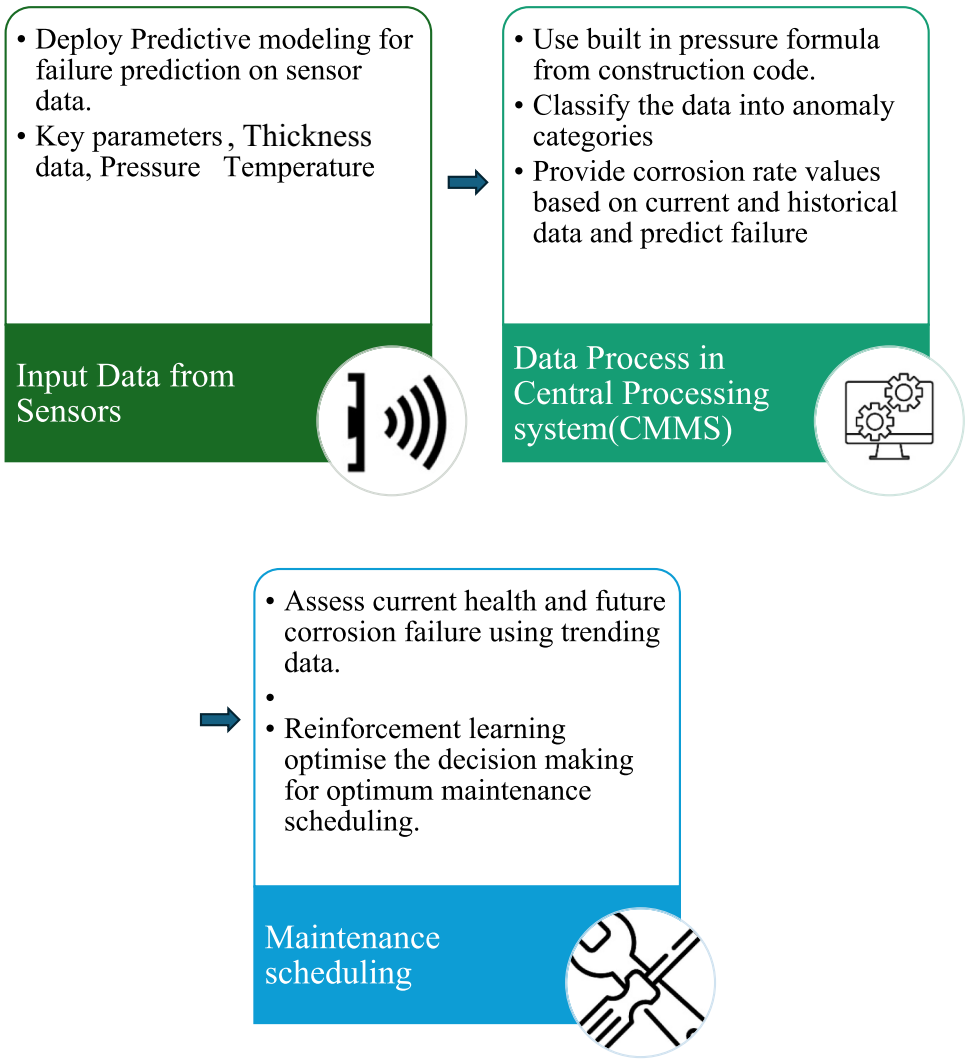


Fig. 17 Internal degradation management work process



Fig. 18 Sample of corrosion defects (Shell Archive files 2019)

This automation approach eliminates the reliance on manual analysis, which is prone to inconsistencies and errors in defect grade classification. Imaging equipment continuously scans the asset, eliminating the need for time-consuming and hazardous manual on-site inspections. In addition, the command center can automatically analyze images and validate

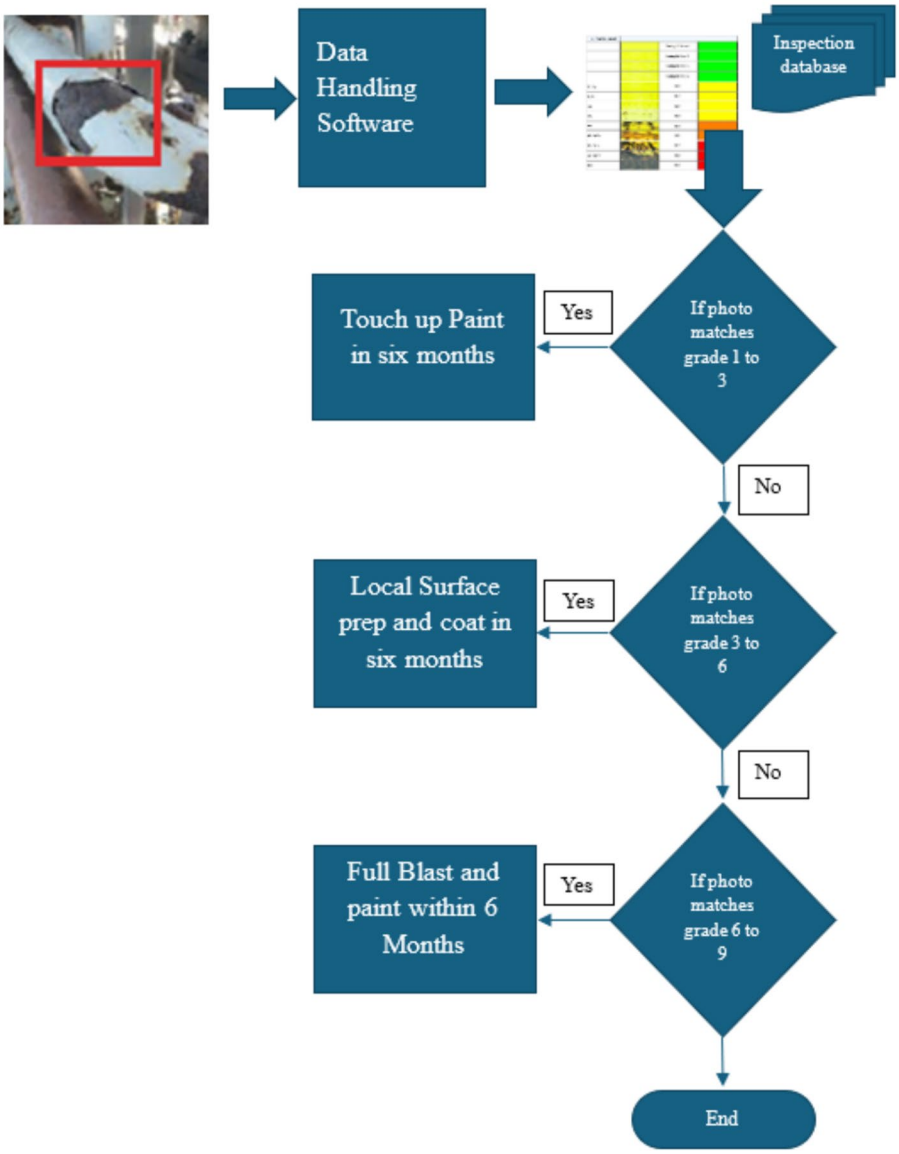


Fig. 19 Typical logic of corrosion evaluation applied in the ML algorithm

samples periodically to ensure that the results generated are accurate.

3.1.3.3 AI Tools and algorithmic strategies for each phase of AIM **Phase 1**: Operational Asset Integrity Management

The objective of this first step is the automation of evaluating the safety critical elements (SCEs), ongoing monitoring of corrosion, and initiating proactive maintenance based on the real-time data stream. These activities are facilitated using a layered AI toolkit. First, this includes embedding supervised and unsupervised machine learning algorithms (Support Vector Machines, Isolation Forests, Random Forests, and deep neural networks, including CNNs and LSTMs) into Computerized Maintenance Management Systems (CMMS) to raise alerts regarding anomalies in wall thickness, pressure, temperature, and corrosion rate signals (Fang et al. 2023; Cui 2024). Second, probabilistic graphical models (Bayesian Networks) capture the interdependencies of AIM factors, providing data-driven failure probabilities to inform risk-aware interventions (Dao 2023; Sattari et al. 2022). Third, unsupervised clustering (K Means, DBSCAN) and dimensionality reduction (PCA) algorithms partition assets with similar behavior, provide visibility of outliers, and streamline high-volume sensor data, thereby enhancing anomaly detection accuracy (Manchanda 2022). Fourth, the use of deep learning paradigms, such as CNN-based vision models and autoencoders, focuses on higher pattern recognition of inspection images and multimodal sensor feeds, resulting in a highly reliable and reproducible corrosion rate measurement (Spencer et al. 2019; Rajendran and Subbian 2025). Lastly, reinforcement learning agents

iteratively learn to cycle inspections and maintenance work orders to minimize lifetime costs under integrity risk constraints(Ogunfowora and Najjaran 2023). In combination, these complementary AI techniques elevate the Phase 1 periodic, manual foundation to a continuous, autonomous, and insight-based integrity assurance regime.

Phase 2: Asset Life Extension (ALE)

This phase utilises a comprehensive set of AI tools to look after the ageing infrastructure, predict the remaining useful life (RUL), prioritize actions or intervention whilst balancing the risk, cost and downtime. Deep Q Networks and Proximal Policy Optimisation are among the reinforcement-learning agents that constantly learn in response to the health indicators so as to optimise the maintenance actions that incur the least lifecycle risks and costs (Pliego Marugán et al. 2024). LSTMs, Temporal Convolutional Networks and Deep Survival Machines are examples of deep time-series models used in the prediction of degradation measures like the loss of wall-thickness and corrosion rate to provide probabilistic RUL curves (Lahiru Jayasinghe 2018). Gaussian-process regression calibrates the bounds of uncertainty in RUL due to sparse data (J. Cui et al. 2024), and physics-informed neural networks allow the incorporation of corrosion or creep physics in the learning procedure, enhancing extrapolation outside the historically observed operating envelopes (Zhang and Shafieezadeh 2023). Graph neural networks are used to identify spatial and functional dependencies across interconnected assets—e.g. complex piping networks enabling life-extension strategies at a holistic level. Real-time machine-learning/finite-element digital twins absorb sensor data, perform ad hoc what-if simulations and dynamically rank life-extension alternatives (Wang et al. 2024) Probabilistic prognostics allows integration of the inline inspection (ILI), SCADA and imagery data into a probabilistic estimate of the wall thickness to come up with stochastic estimates of RUL using Hidden Markov Models and Bayesian hierarchies (Hussain et al. 2024). Generative adversarial networks provide synthetic, realistic degradation paths, complementing degradation data to improve model robustness (He and Zhou 2024). Multi-objective evolutionary optimizers (such as NSGA-II) in conjunction with surrogate ML models obtain cost-versus-risk Pareto-optimal maintenance schedules, and interactive dashboards combine RL recommendations, forecast uncertainties and clustering-based risk maps to support real-time decisions (Seghier Ben et al. 2025).

Phase 3: AI Application in Integrity Management

Phase 3 operationalizes AI for real-time integrity actions, focusing on both internal degradation (wall-loss, pitting, erosion) and external corrosion/fabric maintenance.

Internal degradation management

Various artificial intelligence techniques are working together to address the issue of internal degradation. Among these, ensemble models like Random Forest, XGBoost, LightGBM, and CatBoost are utilized to analyze multi-sensor data, helping to identify wall-thickness anomalies and assess the nature of corrosion (Taiwo et al. 2024). To enhance the detection of pitting, a hybrid pipeline combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks is employed. Initially, the CNN extracts important spatial features from ultrasonic or phased-array images, which are then relayed to an LSTM classifier to recognize time-based trends. This process facilitates earlier and more precise identification of corrosion issues (Akhlaghi et al. 2023). Transformers, including Vision Transformers and Temporal Fusion Transformers, analyze long-term trends in pipe segments and inspection schedules, thus improving predictions of remaining operational life across varying conditions (Lim et al. 2021). A noteworthy approach is self-supervised contrastive learning, utilizing methods like BYOL or SimCLR. This technique taps into vast amounts of unlabeled sensor data to pre-train robust feature encoders, allowing for cost-effective labelling and reducing the risks of overfitting (Chen et al. 2025; Pöppelbaum et al. 2022). Finally, physics-informed generative adversarial networks are used to create synthetic ultrasound images that adhere to the physical laws governing corrosion growth. This not only enhances the diversity of rare defect types in the training data but also strengthens the overall model performance (He and Zhou 2024).

External corrosion (fabric maintenance)

In the case of external corrosion, AI vision is now used to grade and quantify rust almost in real-time. High-resolution CNNs and Efficient Det models rank coating breakdown using ISO 4628-3/ASTM D610 standards (Spencer et al. 2019; Bukhtoyarov et al. 2019), while U-Net++ and DeepLabV3+ produce pixel-level corrosion masks utilized in volumetric metal loss estimates to inform smarter paint schedules. Few-shot prototypical networks learn to recognize new rusting types with fewer than ten photos, and multimodal models combine RGB, thermal-IR, and laser thickness sensors to remain sensitive even in compromised conditions. Additionally, explainable-AI overlays, such as SHAP or Grad-CAM, identify the specific groupings of rust patches that influence each prediction, providing the inspector with an easy-to-audit trace of what caused a given decision to be made (Aboulhosn et al. 2024).

Explainable AI (XAI) serves as a transparency layer in the proposed AIM framework, ensuring that the outputs of black-box models yield trustworthy results for engineers and regulators. When ensemble or deep learning models detect an internal anomaly (e.g., a sudden drop in wall thickness) or classify external rust severity in drone imagery, XAI tools

such as SHAP, Grad-CAM, and counter-factual explanations can clarify why the model triggered: SHAP identifies which sensors and time steps were most influential; Grad-CAM highlights the pixels that led to a specific ISO 4628-3 rating; counter-factual indicate the operating conditions that would most affect remaining useful life (Lavadiya 2021). By embedding these explanations in the CMMS dashboard and archiving them with each work order, AIM teams can quickly verify alerts, direct repair crews to the appropriate locations, maintain complete audit trails, and even gate low-confidence alarms for human review, all while meeting ISO 55000 and regulator expectations for traceable decision-making.

When combined, these tools transform periodic manual surveys into continuous, unattended corrosion assurance, which reduces hibernation time and extends the lifespan of assets.

3.2 Cost–benefit analysis

The economic analysis and cost estimates presented in the study were carried out by an internal evaluation, considering the implementation of this project on a 4-legged drilling offshore platform facility, which has five piping loops and six pressure equipment.196 Sensors are selected and placed on the most susceptible locations based on a risk-based study carried out by assessing the probability and consequence of failure due to the credible damage mechanisms that are prevalent in the operating envelope. Implementing the proposed AI-based integrity management system requires a one-time investment in sensors, imaging equipment, servers, installation, machine learning modeling, and software development, followed by a yearly maintenance cost.

The AI-based system implemented on the platform is estimated to cost around USD 1.3 million over five years, which offers a clear economic benefit compared to traditional inspection methods that are projected to cost about USD 6.22 million over the same period. By implementing the AI-based system, the company can save up to 4.9 million US dollars over a 5-year period. This amounts to a cost savings of approximately 79%. Refer to Fig. 20 for a detailed cost–benefit comparison and Tables 2 and 3 for the cost breakdown for each approach.

Table 2 details the breakdown of the costs for the proposed AI-based systems, summing up all the cost components and maintenance costs for the next 5 years.

Table 2 AI-based system-cost breakdown

S. No.	Components	Count	Cost (US$)
1	Sensors-piping loop	156	198,120
2	Sensors-pressure equipment	40	50,800
3	Server cost		63,500
4	Image-capturing devices	49	124,460
5	Initial set-up cost		25,400
6	ML programming and hosting cost		88,900
7	Total cost		551,180
8	Yearly maintenance cost for 5 years	5 * 150,000	750,000
9	Total cost for 5 years		1,301,180

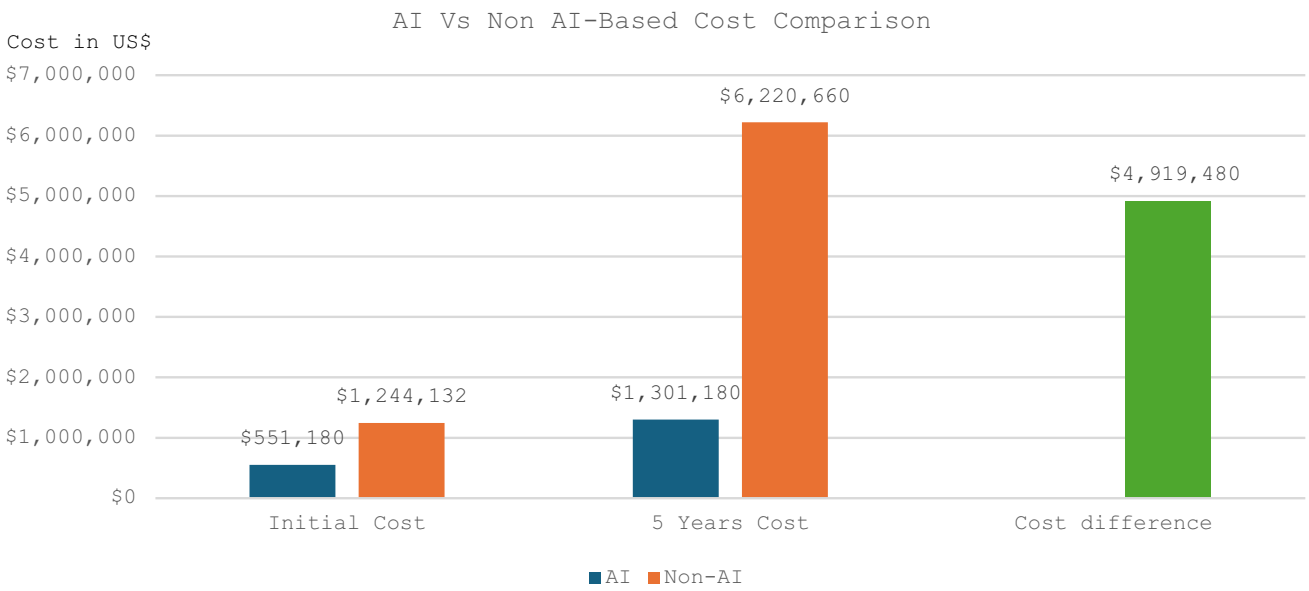


Fig. 20 Cost–benefit analysis

Table 3 Conventional inspection system cost breakdown

S. No.	Components	Count	Cost (US$)
1	Site manhours costs	2	243,332
2	Support staff manhours costs	2	162,267
3	UT equipment cost	4	81,133
4	Logistics and overhead costs		108,200
5	Mobilization and other costs		33,700
6	Engineering and assessment costs	2	488,500
7	Software and hardware costs		127,000
8	Total cost		1,244,132
9	Total cost for 5 years		6,220,660

On the other hand, the non-AI-based system that is currently in use relies on human inspectors visiting offshore locations. This incurs significant manhours, logistics, and equipment costs, which add up to an estimated 6.2 million US dollars over a 5-year period.

While the exact costs can vary depending on the specific asset setup, regional pricing, and vendor rates, the figures used in this study are based on internal evaluations and are in line with costs reported in related industry research and pilot projects (Corrode and Allison 2025; Santamaria and Kinked 2019).

Overall, these findings show that AI-powered integrity systems have the potential to significantly cut lifecycle costs, make inspections more efficient, automate and reduce offshore exposure, all while supporting long-term sustainability goals.

3.3 Limitations and future work

The proposed AI-based AIM framework seems quite promising, but it does have some key limitations to consider. For instance, it depends a lot on high-quality labelled data, which isn't always easy to find. Additionally, its effectiveness may vary in different offshore environments and region-specific factors, such as varying degradation mechanisms. For instance, platforms operating in cold climates might experience issues as icing and brittle fracture, while those operating in tropical environments might be subjected to microbial-induced corrosion. AI Models being developed should also consider environmental exposure factors, material type, expected degradation and operational history to enhance predictive accuracy and relevance.

There's also the continuous need to retrain the model regularly to keep pace with changes in asset conditions. System integration, Regulatory compliance, change management, and training are the other key challenges anticipated in real-time implementation. Acknowledging and addressing these real-world challenges offers a pragmatic roadmap to real-world implementation.

Moving forward, it's important to prioritize the creation of more intuitive transfer learning methods and to improve how AI's decision-making is explained, especially in highly critical situations. More resources need to be dedicated to validating AI systems that ensure integrity in real-life scenarios. Additionally, it's essential to bring together data from different platforms, refine the accuracy of digital twins, and establish standard benchmarks for measuring AI performance in offshore environments. Taking these steps will be vital for this method's progress in the future.

4 Conclusion

Efficient and safe management of offshore oil and gas infrastructure are challenging due to harsh marine environments, complex logistical requirements, and the need for regular inspection and maintenance. Pipelines, being one of the most critical components of offshore oil and gas assets, are susceptible to defects that may not be externally visible but can lead to system-wide failures if not identified and managed in time. This paper explores potential methods for implementing efficient asset management in offshore oil and gas pipelines through AI-based tools and techniques.

Initially, the state of the art in this field was analyzed using Citation Network Analysis (CNA), which identified 177 key scholarly publications related to AI applications in managing oil and gas pipelines. By applying modularity-based clustering to these publications, eight distinct research clusters, each representing a specific thematic area, were identified. This clustering helped trace evolutionary paths, emerging research fronts, critical gaps and directions of future research in AI-assisted asset integrity management of pipelines.

The study provides a structured three-phase AI-based Asset Integrity Management (AIM) framework tailored for offshore oil and gas platforms. While the framework is conceptual in nature, it is based on current technological capabilities and recent research trends. By integrating advanced AI techniques such as supervised learning, reinforcement learning, deep vision models and explainable AI, the proposed approach aims to shift from a reactive approach to a proactive, continuous system. The cost–benefit analysis demonstrates a significant savings of 79% over a 5-year period.

While initial challenges such as data quality, sensor reliability, integrating AI outputs into existing operational workflows and building user confidence through transparent and interpretable models exist, these are expected to be overcome with continued system refinement, cross-disciplinary collaboration and training the workforce on the digital transformation journey. Moreover, region-specific environmental factors and damage mechanisms need to be

considered while developing the models. The findings of this study pave the way for further research and innovation in AI-enabled asset integrity management, with significant implications for improving process efficiency, economy, and safety, while helping extend the operational life of offshore oil and gas pipelines.

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Declarations

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